

Computational Method in Chemical Engineering T2730C2

Textbooks:

- Lorens T. Biegler, E. Ignacio Grossmann, Arthur W. Westerberg, Systematic Methods of Chemical Process Design, Prentice Hall International.
- Warren D. Seider, J. D. Seader, Daniel R. Lewin, Product and Process Design Principles: Synthesis, Analysis, and Evaluation, 2nd Edition, Wiley.
 - Robin Smith, Chemical Process: Design and Integration, Wiley.
- James M. Douglas, Conceptual Design of Chemical Processes, McGraw Hill International, 1988.

Unit 3 – Synthesis of Separation Trains

- Objectives
- Introduction
- Phase Separation of Reactor Effluent
- Criteria for Selection of Separation Methods
- Selection of Equipment
- Sequencing of Ordinary Distillation for the Separation of Nearly Ideal Fluid Mixtures
- Heuristics for Determining Favourable Sequences
- Marginal Vapour Rate Method
- Complex and thermally coupled distillation
- Sequencing of Ordinary Distillation for the Separation of nearly non-Ideal fluid mixtures

Objectives

1. Be familiar with the more widely used industrial separation methods and their basis for separation.
2. Understand the concept of the separation factor and be able to select appropriate separation methods for vapor, liquid, and solid-fluid mixtures.
3. Understand how distillation columns are sequenced and how to apply heuristics to narrow the search for a near-optimal sequence.
4. Be able to apply algorithmic methods to determine an optimal sequence of distillation type separations.
5. Be familiar with the difficulties in and techniques for determining feasible sequences when azeotropes can form.
6. Be able to determine feasible separation systems for gas mixtures and solid-fluid systems.

Introduction

- Almost all chemical processes require separation of mixtures of chemical species.
- A **feed separation system** may be required to purify the reactor feed(s) by removing catalyst poisons and inert species, especially if present as significant percentage of feed.
- An **effluent separation system**, which follows the reactor system and is almost always required, recovers unconverted reactants (in gas, liquid, and/or solid phases) for recycle to the reactor system and separates and purifies products and byproducts.
- Where separations are too difficult, purge streams are used to prevent buildup of certain species in recycle streams.
- Processes that do not involve a reactor system also utilize separation operations if the feed is a mixture that requires separation.
- **Frequently, the major investment and operating costs of a process will be those costs associated with the separation equipment, rather than with the chemical reactors.**

Flowsheet for a process with 1 reactor system

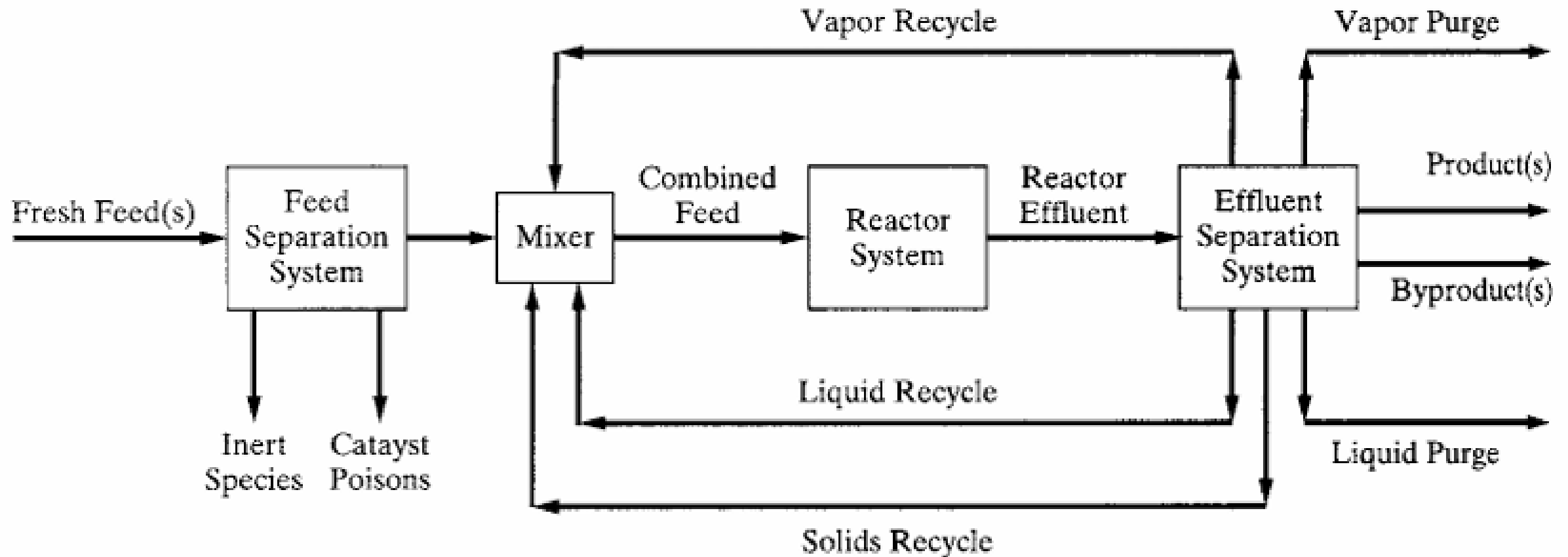


Figure 7.1 General flowsheet for a process with one reactor system.

Phase Separation of Reactor Effluent

- Reactor effluent may be a **heterogeneous** (two or more phases) mixture, but most often is a **homogeneous** (single-phase) mixture.
- For **homogeneous mixtures**, it is advantageous to **change temperature and/or pressure to obtain a partial separation of the components by forming a heterogeneous mixture** of two or more phases.

Possible Phase Conditions of Reactor Effluent

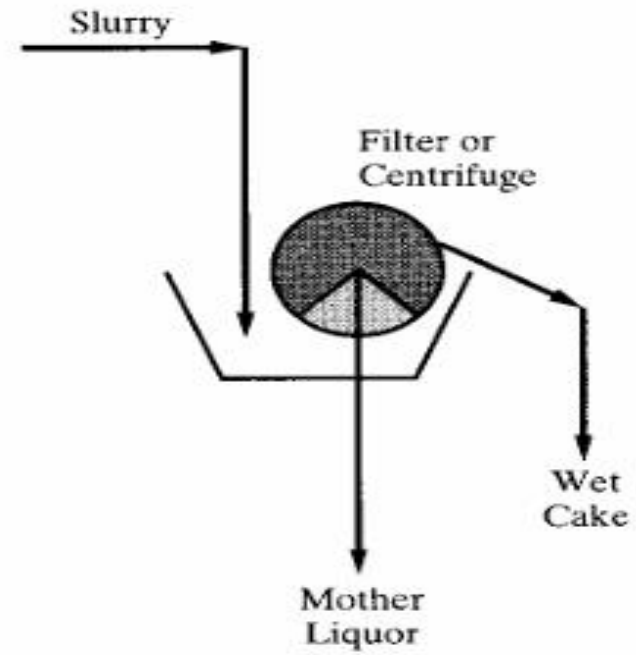
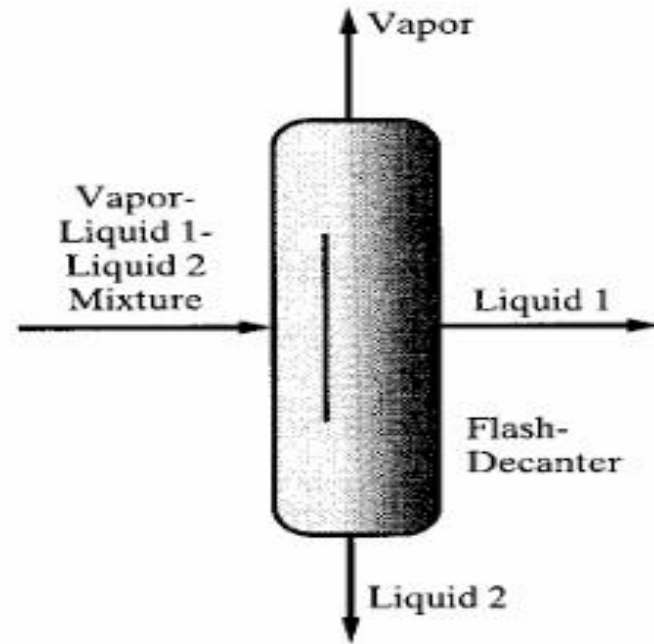
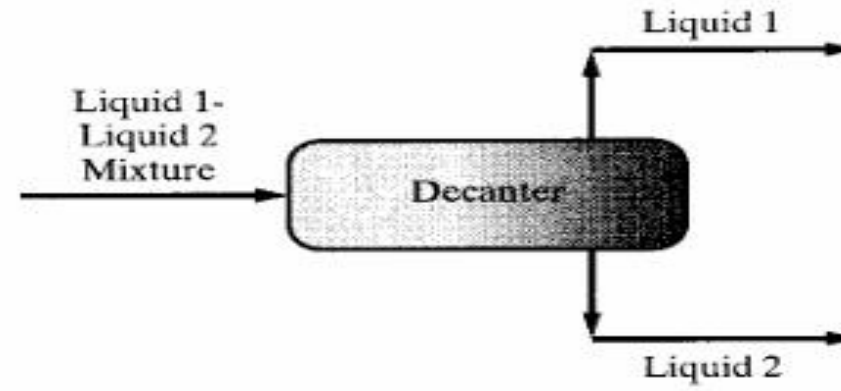
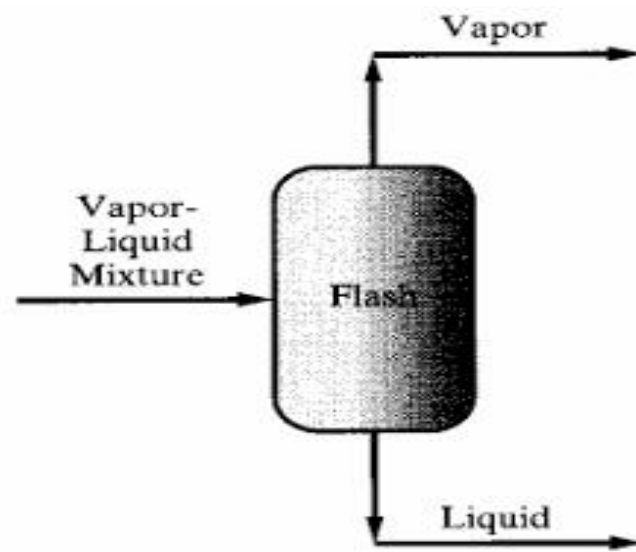
Vapor	Liquid
<i>Vapor and Liquid</i>	<i>Liquid 1 and Liquid 2</i>
<i>Vapor, Liquid 1, and Liquid 2</i>	<i>Liquid and Solids</i>
Vapor and Solids	Liquid 1, Liquid 2, and Solids
Vapor, Liquid, and Solids	Solids
Vapor, Liquid 1, Liquid 2, and Solids	

Phase Separation of Reactor Effluent

- With the reasonable assumption that the **phases in a heterogeneous mixture** are in **phase (physical) equilibrium** for a given reactor effluent composition at the temperature and pressure to which the effluent is brought, **process simulators can readily estimate the amounts and compositions of the phases in equilibrium** by an **isothermal (two-phase)-flash calculation**, provided that solids are not present.
- When the possibility of two liquid phases exists, it is necessary to employ a three-phase flash model, rather than the usual two-phase flash model.
- The three-phase model considers the possibility that a vapor phase may also be present, together with two liquid phases.
- In the absence of solids, the **resulting phases are separated, often by gravity, in a flash vessel for the V-L case, or in a decanter for the V-LI-L2 or LI-L2 cases.**
- For the latter two cases, **centrifugal force** may be employed if gravity settling is too slow because of small liquid-density differences or high liquid viscosities.

Phase Separation of Reactor Effluent

- If solids are present with one or two liquid phases, it is not possible to separate completely the solids from the liquid phase(s).
- Instead, a **centrifuge or filter is used to deliver a wet cake of solids** that requires further processing to recover the liquid and dry the solids.



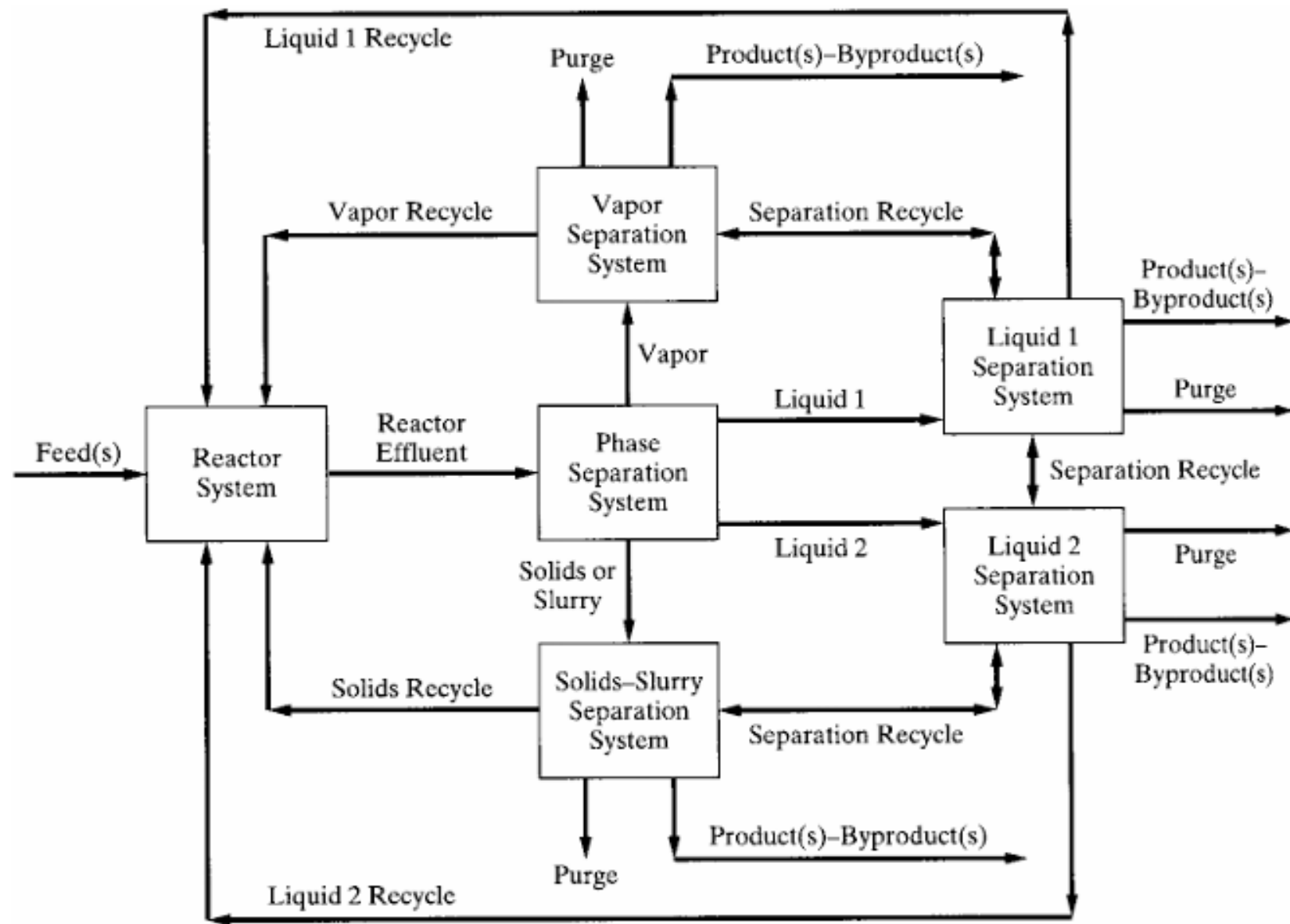


Figure 7.3 Process flowsheet showing separate separation systems with reactor-system and separation-system recycles.

Industrial Separation Processes

- Following phase separation, the individual vapor, liquid, solids, and/or slurry streams are sent to individual separation systems, most common is liquid separation system.
- **When the feed to a vapor or liquid separation system is a binary mixture, it may be possible to select a separation method in just one piece of equipment.**
- For multicomponent separation systems, most separation systems involve a number of units, in which the separations are sequenced, with each unit separating its feed stream into two effluent streams of different composition.
- **The synthesis of a multicomponent separation system can be very complex because it involves not only the selection of the separation methods, but also the manner in which the pieces of separation equipment are sequenced.**
- The separation in each piece of equipment (unit) is made between two components designated as the **key components** for that particular separation unit.
- Each effluent is either a final product or a feed to another separation device.

Industrial Separation Processes - Example

Synthesis of a separation system for the recovery of **butenes** from a C₄ concentrate from the catalytic dehydrogenation of n-butane

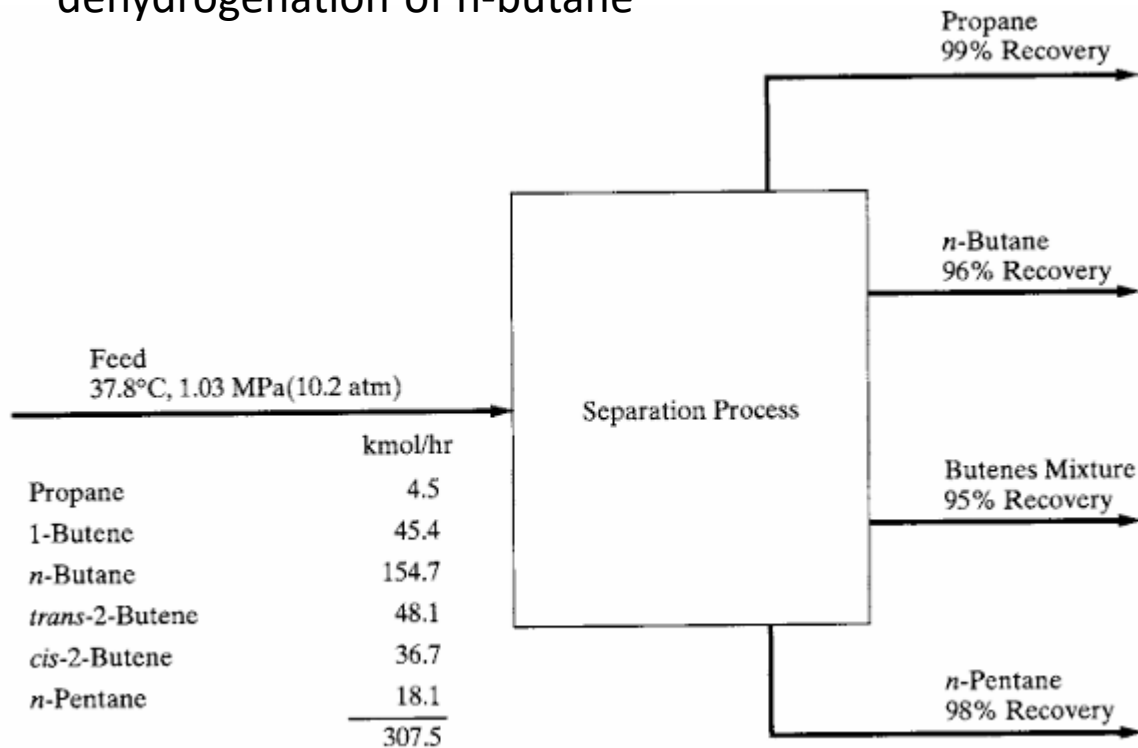


Figure 7.6 Specification for butenes recovery system.

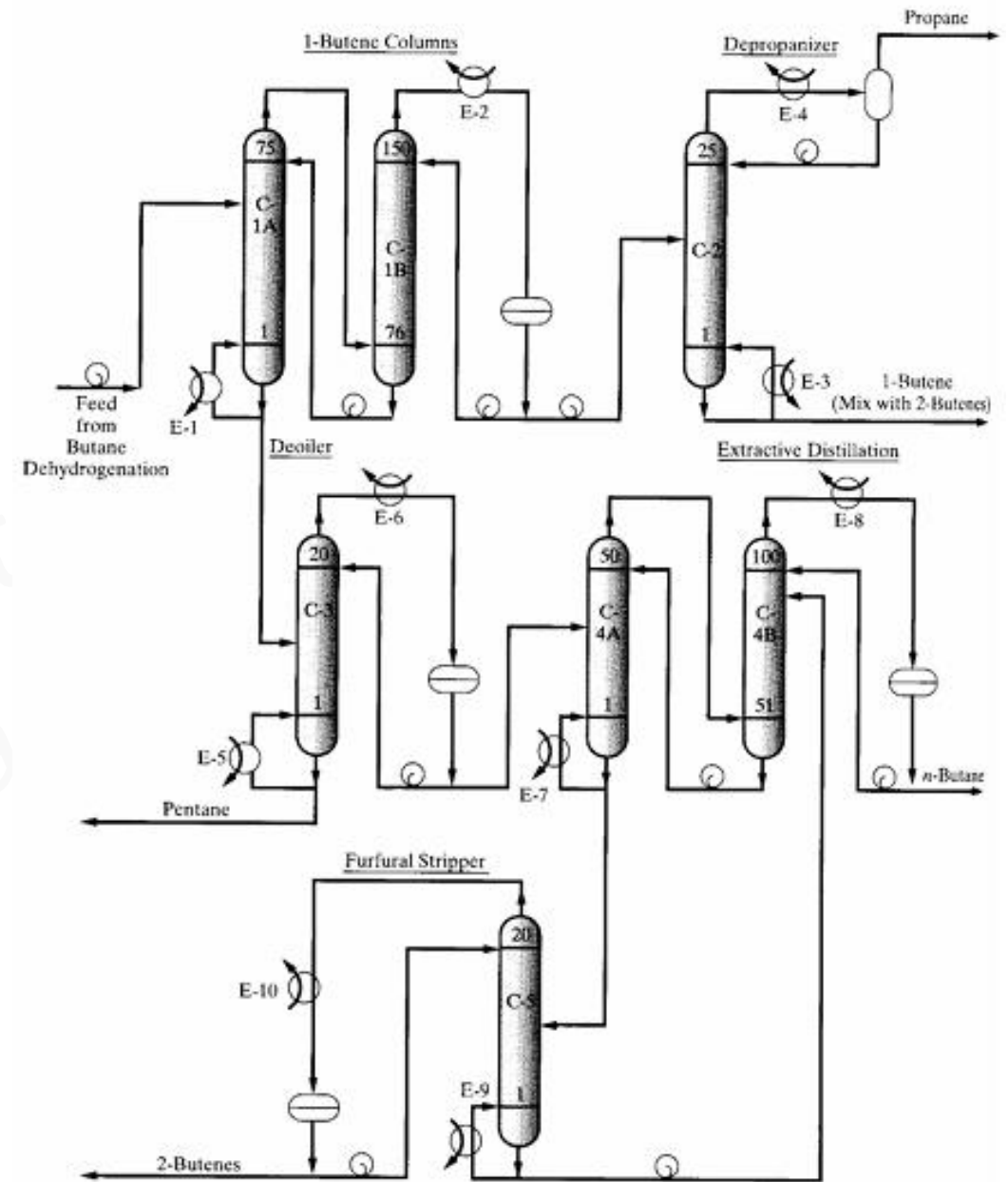


Figure 7.7 Process for butenes recovery: C = distillation column; E = heat exchanger.

Table 7.1 Common Industrial Separation Methods

Separation Method	Phase Condition of Feed	Separating Agent(s)	Developed or Added Phase	Separation Property
Flash	L and/or V	Pressure reduction or heat transfer ESA	V or L	Volatility
Distillation (ordinary)	L and/or V	Heat transfer or shaft work ESA	V or L	Volatility
Gas absorption	V	Liquid absorbent MSA	L	Volatility
Stripping	L	Vapor stripping agent MSA	V	Volatility
Extractive distillation	L and/or V	Liquid solvent and heat transfer MSA	L and V	Volatility
Azeotropic distillation	L and/or V	Liquid entrainer and heat transfer MSA	L and V	Volatility
Liquid-liquid extraction	L	Liquid solvent MSA	Second L	Solubility
Crystallization	L	Heat transfer ESA	S	Solubility or melting point
Gas adsorption	V	Solid adsorbent MSA	S	Adsorbability
Liquid adsorption	L	Solid adsorbent MSA	S	Adsorbability
Membrane	L or V	Membrane ESA	Membrane	Permeability and/or solubility
Supercritical extraction	L or V	Supercritical solvent MSA	Supercritical fluid	Solubility
Leaching	S	Liquid solvent MSA	L	Solubility
Drying	S and L	Heat transfer ESA	V	Volatility
Desublimation	V	Heat transfer ESA	S	Volatility

**MSA – mass
separating agent**

**ESA – energy
separating agent**

Industrial Separation Processes

- The separation of a feed mixture into streams of differing chemical composition is achieved by any one or a combination of four common industrial techniques:
 - 1. Creation by heat transfer, shaft work, or pressure reduction of a second phase**
 - Separation is achieved by an energy-separating agent (ESA), usually heat transfer, which causes formation of a second phase
 - Components are separated by differences in volatility
 - 2. Introduction into the system of a second fluid phase**
 - A second phase is added to the separation unit in the form of a solvent as a mass-separating agent (MSA) that selectively dissolves or alters the volatility of certain species of the mixture
 - Solvent is recycled after another separation unit

Industrial Separation Processes

3. Addition of a solid phase on which selective adsorption can occur

- The addition of solid particles that selectively adsorb certain species of the mixture
- Separation process to recover adsorbent

4. Placement of a selective membrane barrier

- Barrier that allows the permeation of some species over others.
- A mechanical energy loss accompanies the permeation thus technique involves an ESA.

Criteria for Selection of Separation Methods

- **The development of a separation process requires the selection of**
 1. Separation methods
 2. ESAs and/or MSAs
 3. Separation equipment
 4. Optimal arrangement or sequencing of the equipment
 5. Optimal operating conditions of temperature and pressure for the equipment
- For binary mixtures, 1 single separation unit if ESA used, 2 separation units needed if MSA used as MSA needs to be separated and recycled.
- **For multicomponent mixture, minimum number of operation = no. of products – 1.**

Criteria for Selection of Separation Methods

1. Phase Condition of the Feed

i. Vapor Feeds

- If the feed is a vapor or is readily converted to a vapor, operations should be considered: (1) partial condensation, (2) distillation under cryogenic conditions, (3) gas absorption, (4) gas adsorption, (5) gas permeation with a membrane, and (6) desublimation.

ii. Liquid Feeds

- If the feed is a liquid or is readily converted to a liquid, operations applicable: (1) flash or partial vaporization, (2) (ordinary) distillation, (3) stripping, (4) extractive distillation, (5) azeotropic distillation, (6) liquid-liquid extraction, (7) crystallization, (8) liquid adsorption, (9) dialysis, reverse osmosis, ultrafiltration, and pervaporation with a membrane, and (10) supercritical extraction.
- A flash and the different types of distillation are also applicable for feeds consisting of combined liquid and vapor phases.

Criteria for Selection of Separation Methods

1. Phase Condition of the Feed

iii. Slurries, Wet Cakes, and Dry Solids

- Slurry feeds are generally separated first by filtration or centrifugation to obtain a wet cake, which is then separated into a vapor and a dry solid by drying.
- Feeds consisting of dry solids can be leached with a selective solvent to separate the components.

Criteria for Selection of Separation Methods

2. Separation Factor

- **Separation factor SF:** that can be achieved by the particular separation method for the **separation between two key components of the feed.**
- This factor, for the separation of key component 1 from key component 2 between phases I and II, for a single stage of contacting, is defined as

$$SF = \frac{C_1^I/C_2^I}{C_1^{II}/C_2^{II}}$$

- where C_j is a composition (expressed as mole fraction, mass fraction, or concentration) of component j in phase i .
- If phase I is to be rich in component 1 and phase II is to be rich in component 2, then SF must be large.

Criteria for Selection of Separation Methods

2. Separation Factor

- Larger the value of SF, the more feasible is the particular separation operation.
- Operations employing an ESA are economically feasible at a lower value of SF than are those employing an MSA.
- Provided that vapor and liquid phases are readily formed, **distillation should always be considered first as a possible separation operation if the feed is a liquid or partially vaporized.**
- However, when seeking a desirable SF value, it is best to avoid extreme conditions of temperature that may require refrigeration or damage heat-sensitive materials; pressures that may require gas compression or vacuum; and MSA concentrations that may require expensive means to recover the MSA.

Separation Process	Separation Mechanism	Separation Factor
Distillation	Relative volatility α	$SF > 1$ $SF = \frac{y_1/y_2}{x_1/x_2} = \frac{y_1/x_1}{y_2/x_2} = \frac{K_1}{K_2} = \alpha_{1,2}$
Multicomponent liquid vapor separation operations	ESA (partial evaporation, partial condensation, or distillation)	P_i^s is the vapor pressure of component i $SF = \alpha_{1,2} = \frac{P_1^s}{P_2^s}$
Non-ideal liquid solutions	Equation-of-state (SRK or PR)	$\bar{\Phi}_i$ is mixture fugacity coefficient of component i $SF = \alpha_{1,2} = \frac{\bar{\Phi}_1^L/\bar{\Phi}_1^V}{\bar{\Phi}_2^L/\bar{\Phi}_2^V}$
Azeotropic and extractive distillation that use an MSA that forms a nonideal liquid solution		γ_i is liquid-phase activity coefficients $SF = \alpha_{1,2} = \frac{\gamma_1^L P_1^s}{\gamma_2^L P_2^s}$
Liquid-liquid extraction	MSA is used to create two liquid phases	Relative selectivity $SF = \beta_{1,2} = \frac{\gamma_1^{II}/\gamma_2^{II}}{\gamma_1^I/\gamma_2^I}$
Membrane separation process	P_{Mi} is the permeability of species i recovery factor RF: degree of sharpness of separation	$SF = \frac{P_{M_1}}{P_{M_2}}$ $RF = \frac{n_i^I}{n_i^F}$

Criteria for Selection of Separation Methods

3. Reason for the Separation as a Criterion

- A final consideration in the selection of a separation method is the reason for the separation.
- **Possible reasons are (1) purification of a species or group of species, (2) removal of undesirable constituents, and (3) recovery of constituents for subsequent processing or removal.**
- In the case of purification, the use of an MSA method may avoid exposure with an ESA method to high temperatures that may cause decomposition.
- In some cases, removal of undesirable species together with a modest amount of desirable species may be economically acceptable.
- Likewise, in the recovery of constituents for recycle, a high degree of separation from the product(s) may not be necessary.

Selection of Equipment

Absorption, Stripping, and Distillation

- Either trayed or packed columns are used.
- Trayed columns are usually preferred for initial installations, columns ≥ 3 ft in diameter.
- Trayed columns for feeds containing solids, high liquid-to-gas ratios, large-diameter columns, and where operation over a wide range of conditions is necessary.
- Commonly used tray types are sieve, valve, and bubble-cap.
- Bubble-cap tray has higher cost, used only when a large liquid holdup is required tray, e.g. when conducting a chemical reaction simultaneously with distillation.
- Sieve trays least expensive and have lowest pressure drop per tray, but they have the narrowest operating range (turndown ratio).

Selection of Equipment

Absorption, Stripping, and Distillation

- When flexibility is required, valve trays are a better choice.
- Packed columns for operation under vacuum or where a low-pressure drop desired, corrosive systems, foaming systems, and cases where low liquid holdup is desired.
- Different types of packings available are random or structured.
- Structured packing is more expensive but has lower pressure drop, higher efficiency, and higher capacity compared to both random packings and trays.
- Structured packings are often considered for column revamps.

Selection of Equipment

Liquid-Liquid Extraction

- Variety of equipment including multiple mixer-settler units or single countercurrent-flow columns with or without mechanical agitation, centrifugal extractors (compact, expensive).
- Mixer-settler units used when the equivalent of a few theoretical stages required, efficiencies approach 100%.
- For a large number of stages, columns with mechanical agitation may be favored.
- Packed and perforated tray columns can be very inefficient and are not recommended for critical separations.

Selection of Equipment

Membrane Separation

- Use natural or synthetic, glassy or rubbery polymers.
- For high permeability and selectivity, nonporous materials are preferred, with thicknesses ranging from 0.1 to 1.0 micron, either as a surface layer or film onto or as part of much thicker asymmetric or composite membrane materials
- Materials are fabricated primarily into spiral-wound and hollow-fiber-type modules to achieve a high ratio of membrane surface area to module volume.

Selection of Equipment

Absorption

- Adsorbent must be chosen carefully to give the required selectivity, capacity, stability, strength, and regenerability.
- Commonly used adsorbents are activated carbon, molecular-sieve carbon, molecular-sieve zeolites, silica gel, and activated alumina.
- Particular importance in the selection process is the adsorption isotherm for competing solutes when using a particular adsorbent.
- Most adsorption operations are conducted in a semicontinuous cyclic mode that includes a regeneration step.
- Batch slurry systems are favored for small-scale separations, whereas fixed-bed operations are preferred for large-scale separations.

Selection of Equipment

Leaching

- Equipment for leaching operations is designed for either batchwise or continuous processing.
- For rapid leaching, it is best to reduce the size of the solids by grinding or slicing.
- The solids are contacted by the solvent using either percolation or immersion.

Crystallization

- Crystallization operations include the crystallization of an inorganic compound from an aqueous solution (solution crystallization) and the crystallization of an organic compound from a mixture of organic chemicals (melt crystallization).
- On a large scale, solution crystallization is frequently conducted continuously in a vacuum evaporating draft-tube baffled crystallizer to produce crystalline particles, whereas the falling-film crystallizer is used for melt crystallization to produce a dense layer of crystals.

Selection of Equipment

Drying

- Factors: nature of the feed (granular solids, a paste, a slab, a film, a slurry, or a liquid), need for agitation, type of heat source (convection, radiation, conduction, or microwave heating), and degree to which the material must be dried.
- Most commonly employed continuous dryers include tunnel, belt, band, turbo-tray, rotary, steam-tube rotary, screw-conveyor, fluidized-bed, spouted-bed, pneumatic-conveyor, spray, and drum dryers.

Operation	Equipment
Absorption, Stripping, and Distillation	Tray or packed column Trayed columns (sieve/valve/bubble cap): feeds containing solids, high liquid-to-gas ratios, large-diameter columns, wide range of conditions Packed columns: vacuum or low-pressure drop desired, corrosive systems, foaming systems, low liquid holdup desired
Liquid-Liquid Extraction	Multiple mixer-settler units (few theoretical stages) Single countercurrent-flow columns with or without mechanical agitation (higher stages) Centrifugal extractors (compact and expensive)
Membrane Separation	Spiral-wound and hollow-fiber-type modules
Adsorption	Batch slurry systems (small-scale separations) Fixed-bed operations are (large-scale separations)
Leaching	Batchwise or continuous processing
Crystallization	vacuum evaporating draft-tube baffled crystallizer (crystalline particles) falling-film crystallizer (melt crystallization, dense layer of crystals)
Drying	Tunnel, belt, band, turbo-tray, rotary, steam-tube rotary, screw-conveyor, fluidized-bed spouted-bed, pneumatic-conveyor, spray, and drum dryers.

Sequencing of Ordinary Distillation for the Separation of Nearly Ideal Fluid Mixtures

- For **nearly ideal feeds**, economical sequence will often include only **ordinary distillation columns** if the conditions hold:
 1. The relative volatility between the two selected key components for the separation in each column is > 1.05 .
 2. The reboiler duty is not excessive.
 3. The tower pressure does not cause the mixture to approach its critical temperature.
 4. The overhead vapor can be at least partially condensed at the column pressure to provide reflux without excessive refrigeration requirements.
 5. The bottoms temperature at the tower pressure is not so high that chemical decomposition occurs.
 6. Azeotropes do not prevent the desired separation.
 7. Column pressure drop is tolerable, particularly if operation is under vacuum.

Sequencing of Ordinary Distillation for the Separation of Nearly Ideal Fluid Mixtures

1. Column Pressure and Type of Condenser

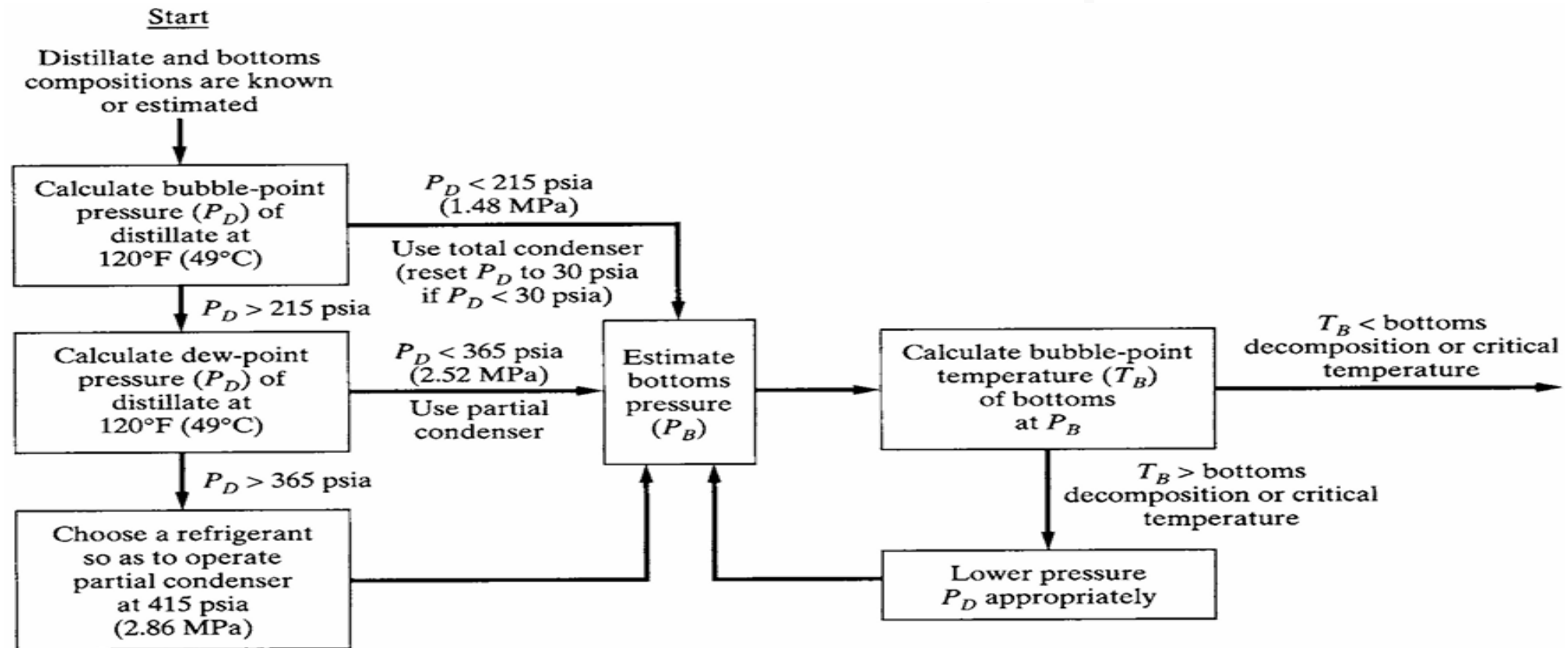


Figure 7.9 Algorithm for establishing distillation column pressure and condenser type.

Sequencing of Ordinary Distillation for the Separation of Nearly Ideal Fluid Mixtures

2. Number of Sequences of Ordinary Distillation Columns

- Initial consideration is usually given to a sequence of ordinary distillation columns, where a single feed is sent to each column and the products from each column number are two, the distillate and the bottoms.
- It takes a sequence of two ordinary distillation columns to separate a mixture into three products.
- **Synthesis of ordinary distillation sequences for a multicomponent feed that is to be separated into P final products (pure components and/or multicomponent mixtures).**
- ***The components in the feed are ordered by volatility***, with the first component being the most volatile, consistent with normal boiling point (if mixture forms nearly ideal liquid solutions).

Sequencing of Ordinary Distillation for the Separation of Nearly Ideal Fluid Mixtures

- An equation for the number of different sequences of ordinary distillation columns, N_s , to produce a number of products, P

$$N_s = \sum_{j=1}^{P-1} N_j N_{P-j} = \frac{[2(P-1)]!}{P!(P-1)!}$$

- For the first separator in the sequence, $P - 1$ separation points are possible.
- Now let j be the number of final products that must be developed from the distillate of the first column
- Then $P-j$ equals the number of final products that must be developed from the bottoms of the first column.
- If N_i is the number of different sequences for i final products, then, for a given separation point in the first column, the number of sequences is $N_j N_{P-j}$.

Sequencing of Ordinary Distillation for the Separation of Nearly Ideal Fluid Mixtures

Table 7.2 Number of Possible Sequences for Separation by Ordinary Distillation

Number of Products, P	Number of Separators in the Sequence	Number of Different Sequences, N_s
2	1	1
3	2	2
4	3	5
5	4	14
6	5	42
7	6	132
8	7	429
9	8	1,430
10	9	4,862

EXAMPLE 7.1

Ordinary distillation is to be used to separate the ordered mixture $C_2, C_3^-, C_3, 1-C_4^-, nC_4$ into the three products C_2 ; $(C_3^-, 1-C_4^-)$; (C_3, nC_4) . Determine the number of possible sequences.

SOLUTION

Neither multicomponent product contains adjacent components in the ordered list. Therefore, the mixture must be completely separated with subsequent blending to produce the $(C_3^-, 1-C_4^-)$ and (C_3, nC_4) products. Thus, from Table 7.2 with P taken as 5, $N_s = 14$. ■

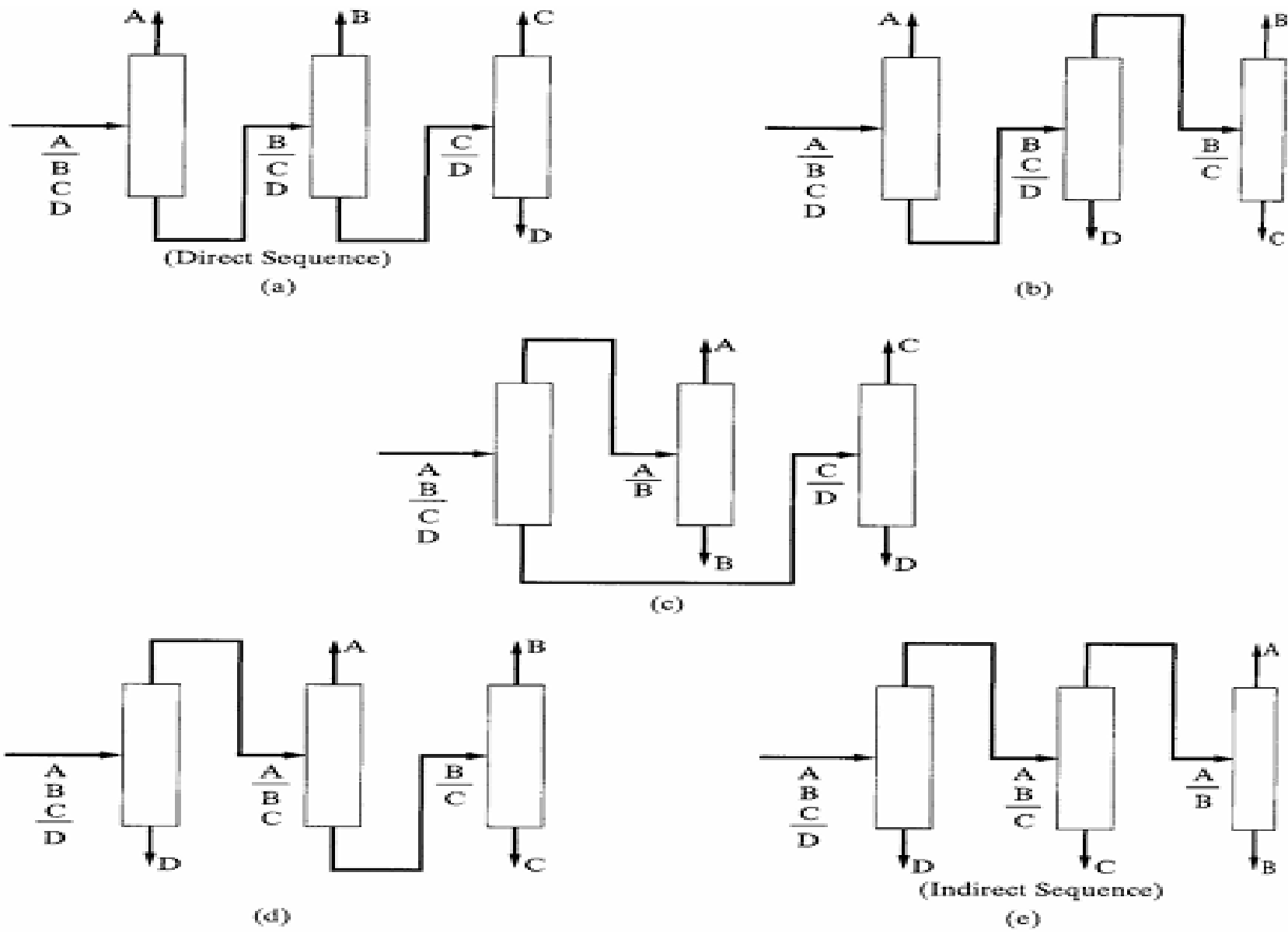


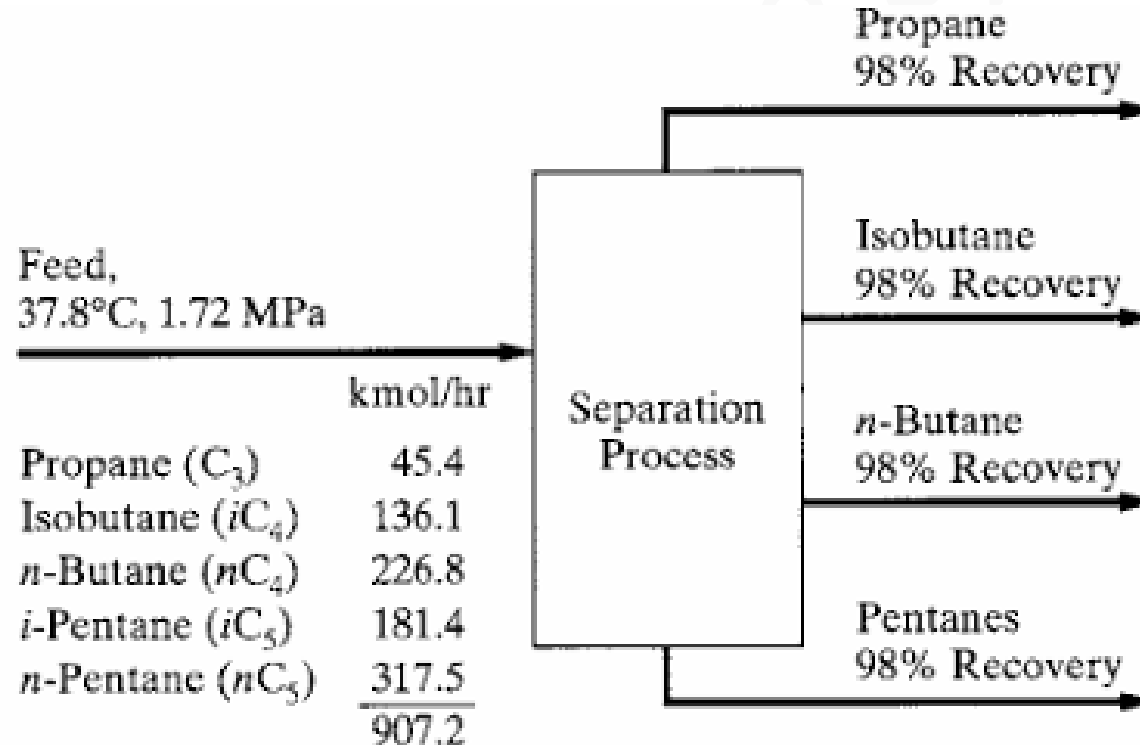
Figure 7.11 The five sequences for a four-component feed.

Heuristics for Determining Favourable Sequences

1. Remove thermally unstable, corrosive, or chemically reactive components early in the sequence.
2. Remove final products one by one as distillates (the direct sequence).
3. Sequence separation points to remove, early in the sequence, those components of greatest molar percentage in the feed.
4. Sequence separation points in the order of decreasing relative volatility so that the most difficult splits are made in the absence of the other components.
5. Sequence separation points to leave last those separations that give the highest-purity products.
6. Sequence separation points that favor near equimolar amounts of distillate and bottoms in each column (cost effective).

Example 7.2

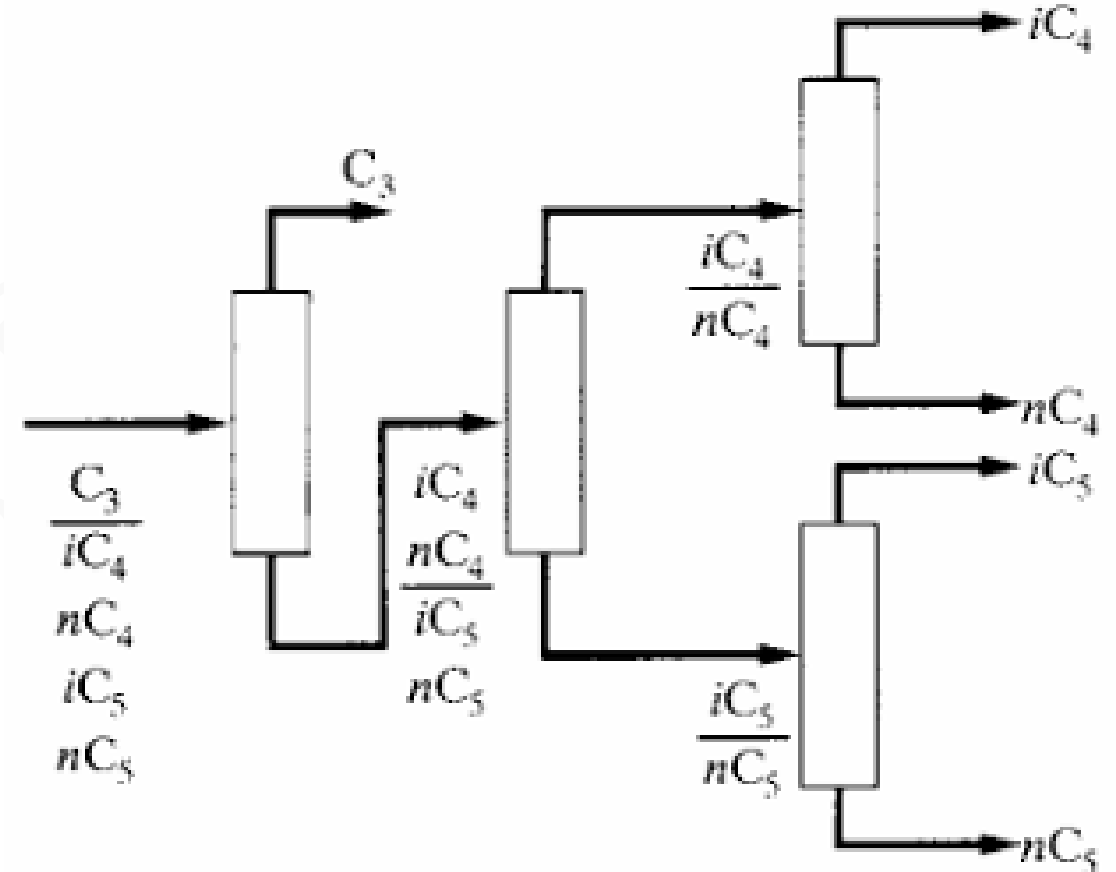
Consider the separation problem shown in Figure 7.12a, except that separate isopentane and *n*-pentane products are also to be obtained with 98% recoveries. Use heuristics to determine a good sequence of ordinary distillation units.



Example 7.2

Approximate relative volatilities for all adjacent pairs

Component pair	Approximate α at 1 atm
C_3/iC_4	3.6
iC_4/nC_4	1.5
nC_4/iC_5	2.8
iC_5/nC_5	1.35



- Wide variations in both relative volatility and molar percentages in the process feed.
- The choice is Heuristic 4, which dominates over Heuristic 3.
- First split is between the pair with the highest relative volatility.
- This sequence also corresponds to the optimal arrangement.

Marginal Vapour Rate Method

- The marginal vapor rate (MV) method of Modi and Westerberg (1992).
- Outperforms the other methods and can be applied without the necessity of complete column designs and calculations of costs.
- For a given split between two key components, the **difference in costs between the separation in the absence of nonkey components and the separation in the presence of nonkey components is the marginal annualized cost (MAC).**
- They show that a **good approximation of MAC is the MV**, which is the corresponding **difference in molar vapor rate passing up the column.**
- ***The sequence with the minimum sum of column MVs is selected.***
- The good approximation is due to the fact that vapor rate is a good measure of cost because it is a major factor in **determining column diameter as well as reboiler and condenser areas** (thus, column and heat exchanger **capital costs**) and reboiler and condenser duties (thus, heat exchanger annual **operating costs**).

Marginal Vapour Rate Method

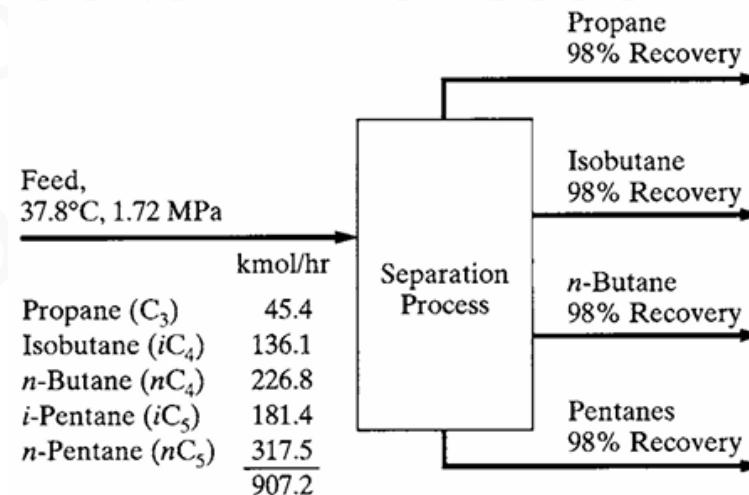
- A convenient method for determining the molar vapor rate in an ordinary distillation column separating a nearly ideal system uses the Underwood equations to calculate the minimum reflux ratio, $R_{\min}$.
- The design reflux ratio is taken as $R = 1.2 R_{\min}$.
- By material balance, the molar vapor rate, V , entering the condenser is given by

$$V = D(R + 1)$$

- where D is the molar distillate rate.
- Assuming that the feed to the column is a bubble-point liquid, the molar vapor rate through the column will be nearly constant at this value of V .
- To simplify the material balance calculations, it is convenient to assume nearly perfect separations with the light and lighter-than-light key components leaving in the distillate and the heavy and heavier-than-heavy key components leaving in the bottoms.
- $R_{\min}$ not affected by purity.
- The column feed pressure is taken as the average of the top and bottom pressures.

Example 7.3

Use the marginal vapor rate (MV) method to determine a sequence for the separation of light hydrocarbons specified in Figure 7.12a, except: (1) remove the propane from the feed, (2) ignore the given temperature and pressure of the feed, and (3) strive for recoveries of 99.9% of the key components in each column. Use a process simulation program, with the Soave–Redlich–Kwong equation of state for K -values and enthalpies, to set top and bottom column pressures and estimate the reflux ratio with the Underwood equations.



Example 7.3

- To produce four nearly pure products from the four-component feed, five sequences of three ordinary distillation columns each are obtained.
- Let A = isobutane, B = n-butane, C = isopentane, and D = n-pentane.
- A total of 10 unique separations are listed.
- Calculations for the top column pressure, P''/P' in kPa; the molar distillate rate, D , in kmol/hr; and the reflux ratio, R , using the shortcut (Fenske-Underwood-Gilliland or FUG) distillation model of the CHEMCAD process simulation program shown in Table 7.3.
- This model applies the Underwood equations to estimate the minimum reflux ratio.
- Column feeds were computed as bubble-point liquids at $P_{\text{top}} = P + 35$ kPa, values of the column molar vapor rate, V , in kmol/hr and marginal vapor rate, MV , in kmol/hr also estimated.
- Calculate the sum of the marginal vapor rates is calculated for each of the five sequences.
- Table 7.4 shows that the preferred sequence is the one that performs the two most difficult separations, A/B and C/D, in the absence of nonkey components.
- These two separations are far more difficult than the separation, B/C. The direct sequence is the next best.

Example 7.3

Table 7.3 Calculations of Marginal Vapor Rate, MV

Separation	Column Top Pressure (kPa)	Distillate Rate, D (kmol/hr)	Reflux Ratio ($R = 1.2 R_{\min}$)	Vapor Rate, $V = D(R + 1)$ (kmol/hr)	Marginal Vapor Rate (kmol/hr)
A/B	680	136.2	10.7	1,594	0
A/BC	680	136.2	11.9	1,757	163
A/BCD	680	136.2	13.2	1,934	340
B/C	490	226.8	2.06	694	0
AB/C	560	362.9	1.55	925	231
B/CD	490	226.8	3.06	921	227
AB/CD	560	362.9	2.11	1,129	435
C/D	210	181.5	13.5	2,632	0
BC/D	350	408.3	6.39	3,017	385
ABC/D	430	544.4	4.96	3,245	613

Calculate total MV for each sequence:

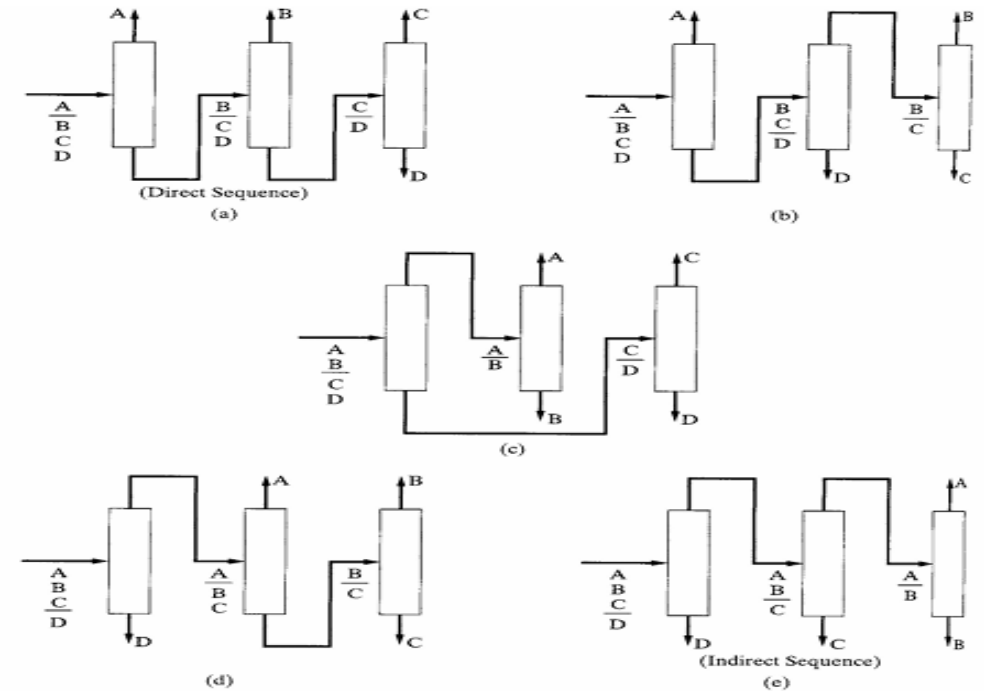


Figure 7.11 The five sequences for a four-component feed.

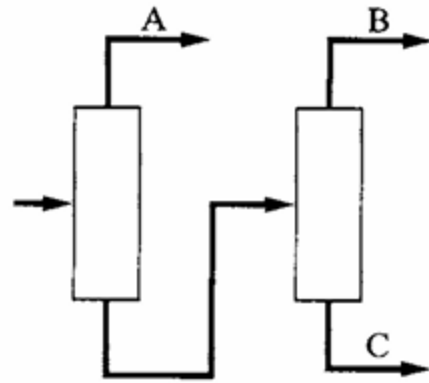
Table 7.4 Marginal Vapor Rates for the Five Possible Sequences

Sequence in Figure 7.11	Marginal Vapor Rate, MV (kmol/hr)
(a) Direct	567
(b)	725
(c)	435
(d)	776
(e) Indirect	844

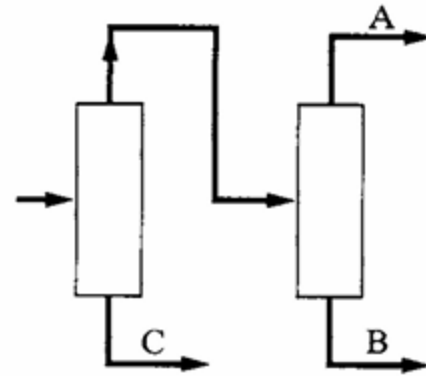
Complex and thermally coupled distillation

- Guide from a **study by Tedder and Rudd (1978)** of the separation of ternary mixtures (A, B, and C in order of decreasing volatility)
- **Eight alternative sequences of one to three columns were considered**, seven of are shown in Figure 7.13.
- The configurations include the direct and indirect sequences (I and II), two interlinked cases (III and IV), five cases that include the use of side streams (III, IV, V, VI, and VII), and one case (V) involving a column with two feeds.
- All columns in Cases I, II, V, VI, and VII have condensers and reboilers.
- In Cases III and IV, the first column has a condenser and reboiler.
- In Case III, the rectifier column has a condenser only, while the stripper in Case IV has a reboiler only.
- The interlinking streams that return from the second column to the first column thermally couple the columns in Cases III and IV.

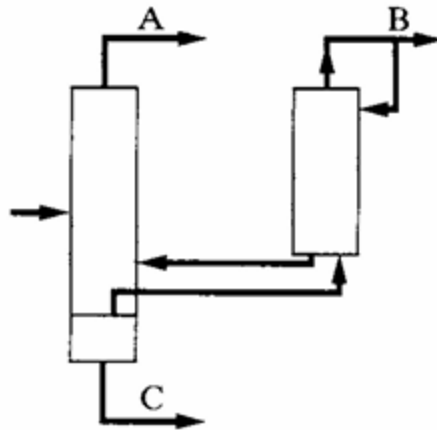
Complex and thermally coupled distillation



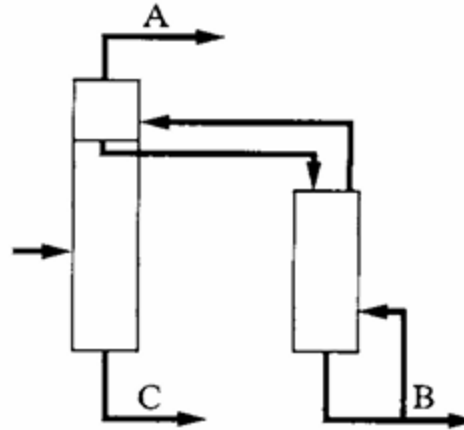
I. Direct Sequence



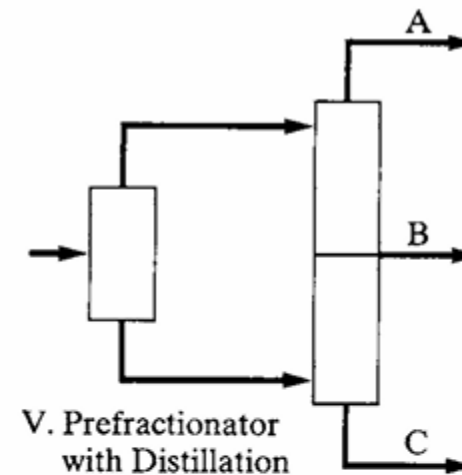
II. Indirect Sequence



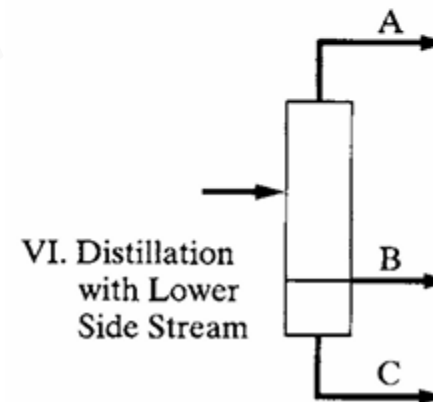
III. Distillation with Vapor Side Stream Rectifier



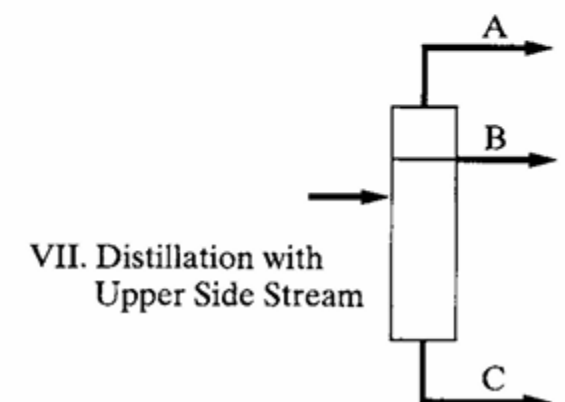
IV. Distillation with Liquid Side Stream Stripper



V. Prefractionator with Distillation



VI. Distillation with Lower Side Stream



VII. Distillation with Upper Side Stream

Figure 7.13 Configurations for ternary distillation.

Complex and thermally coupled distillation

- Optimal regions for the various configurations depend on the **process feed composition** and on an **ease-of-separation index (ESI)**, which is defined as the **relative volatility ratio**.
- Ternary mixture is separated into three products with just one column in Cases VI and VII, but the reflux requirement is excessive unless the feed contains a large amount of B (component of intermediate volatility) and little of the other component removed from the same section of the column as B.
- Otherwise, if the feed is dominated by B but also contains appreciable amounts of A and C, the prefractionator case (V) is optimal.
- Surprise of the study is the superiority of distillation with a vapor sidestream rectifier, which is favored for a large region of the feed composition when $ESI > 1.6$.
- The results of Figure 7.14 can be extended to multicomponent separation problems involving more than three components, if difficult ternary separations are performed last.

Complex and thermally coupled distillation

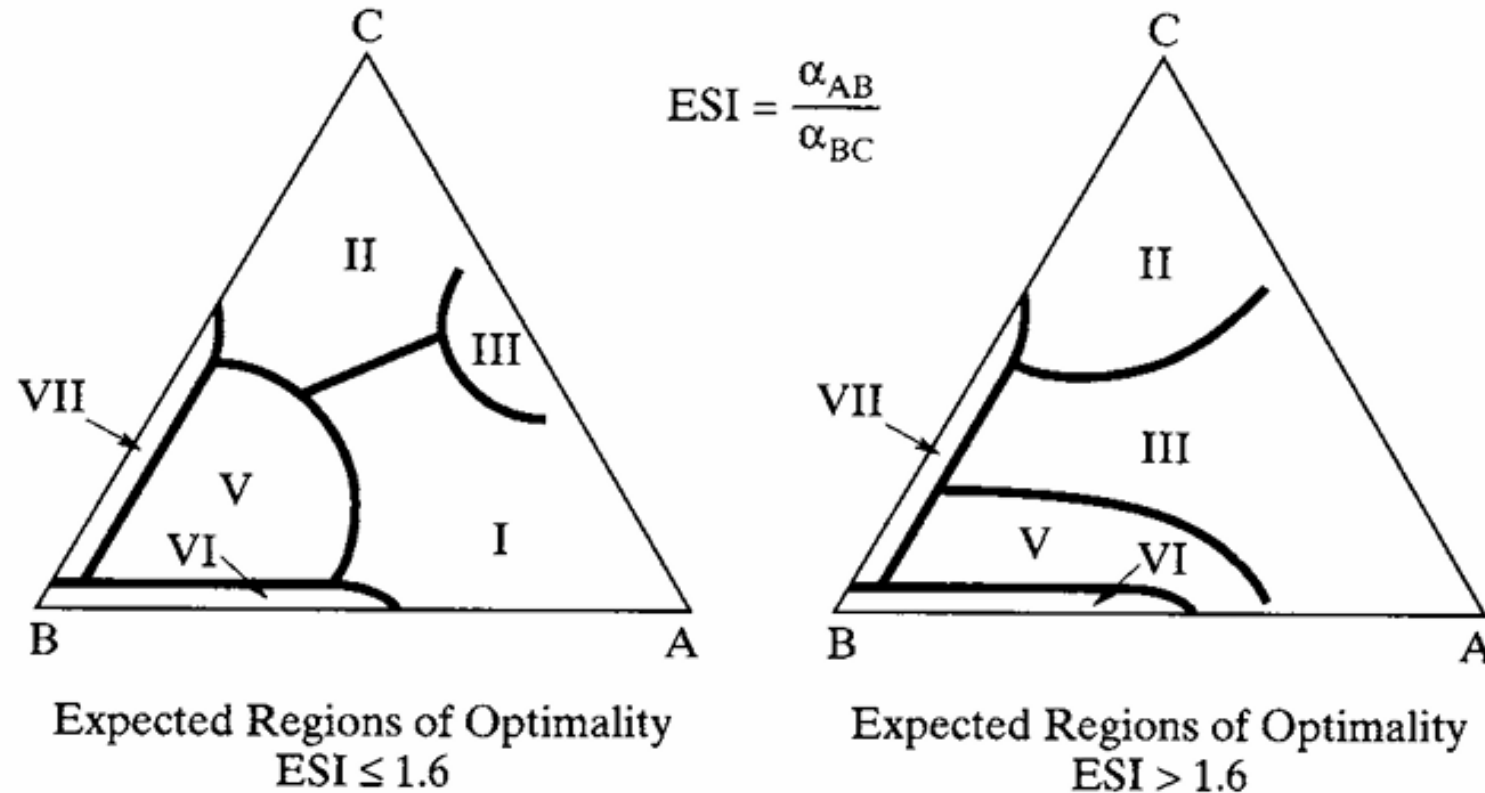


Figure 7.14 Regions of optimality for ternary distillation configurations (Tedder and Rudd, 1978a, b).

Sequencing of Ordinary Distillation for the Separation of nearly non-Ideal fluid mixtures

- When a multicomponent fluid mixture is nonideal, its separation by a sequence of ordinary distillation columns **will not be technically and/or economically feasible** if **relative volatilities between key components drop below 1.05** and, particularly, if **azeotropes are formed**.
- For such mixtures, separation is most commonly achieved by sequences comprised of ordinary distillation columns, enhanced distillation columns, and/or liquid-liquid extraction equipment.
- Membrane and adsorption separations can also be incorporated into separation sequences, but their use is much less common.
- Enhanced distillation operations include extractive distillation, homogeneous azeotropic distillation, heterogeneous azeotropic distillation, pressure-swing distillation, and reactive distillation.

Sequencing of Ordinary Distillation for the Separation of nearly non-Ideal fluid mixtures

Azeotrope:

- **Azeotrope** is an ancient Greek word that is translated "**to boil unchanged**" meaning that the vapor emitted has the same composition as the liquid.
- **The azeotropic composition is commonly referred to as a distillation boundary and pinch point** (equilibrium constant $K = y_i/x_i = 1$).
- When classifying the many azeotropic mixtures, it is helpful to examine their deviations from Raoult's law.

- For ideal mixtures following Raoult's law: $y_1 P = x_1 P_1^s \quad y_2 P = x_2 P_2^s$

$$(y_1 + y_2) P = P = x_1 P_1^s + x_2 P_2^s = x_1 P_1^s + (1 - x_1) P_2^s = P_2^s + (P_1^s - P_2^s)x_1$$

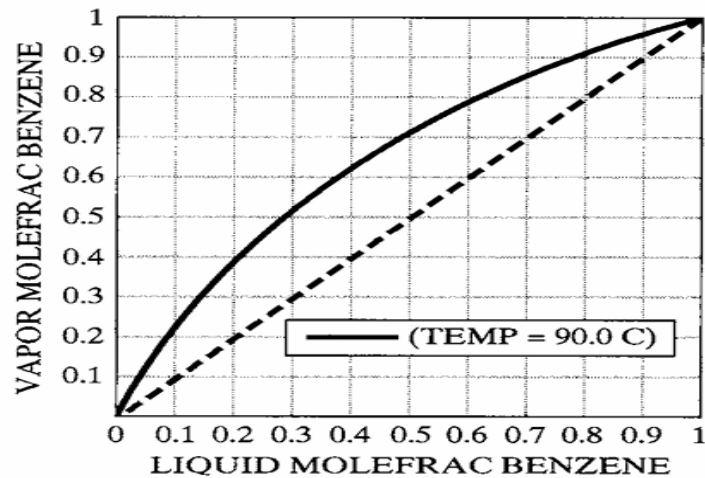
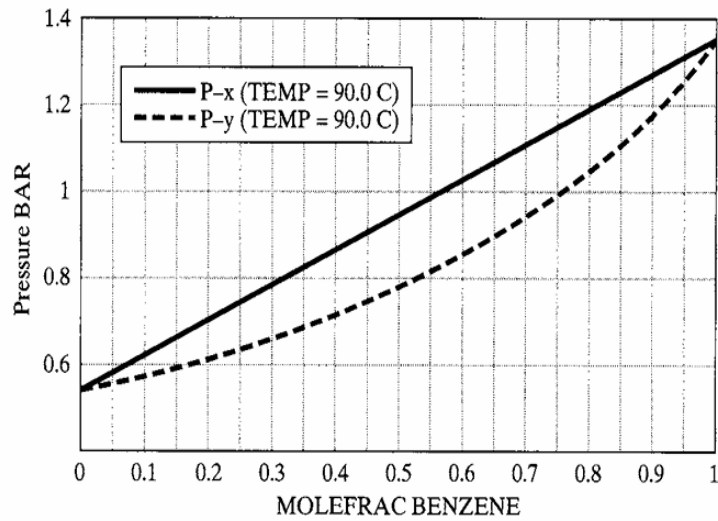
- However, when the mixture forms a nonideal liquid phase and exhibits a positive deviation from Raoult's law

$$y_j \bar{\phi}_j^V P = x_j \gamma_j^L f_j^L \quad (\gamma_j^L > 1, j = 1, 2) \quad P = x_1 \gamma_1^L P_1^s + (1 - x_1) \gamma_2^L P_2^s$$

Non-Ideal Mixture

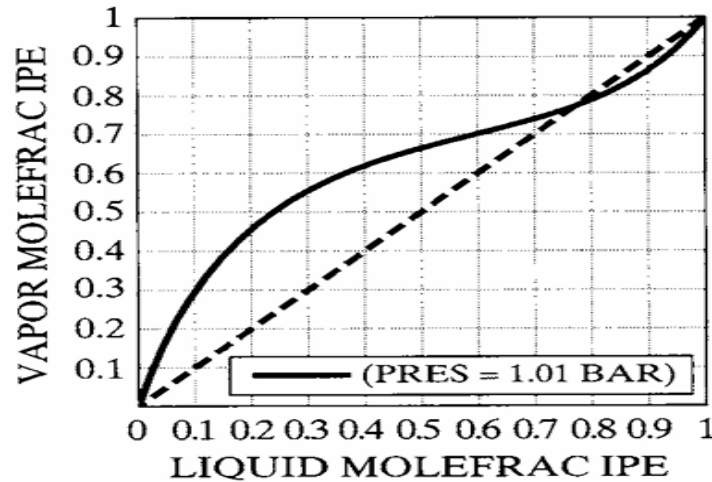
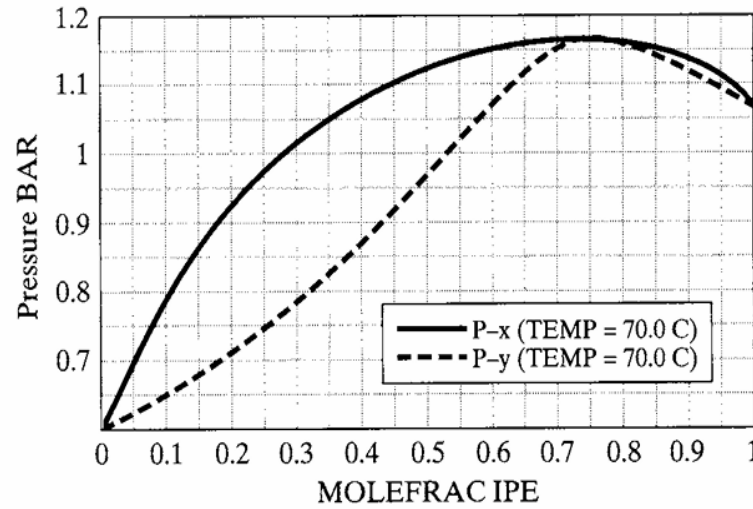
Ideal Mixture

Benzene-Toluene mixture at 90 °C



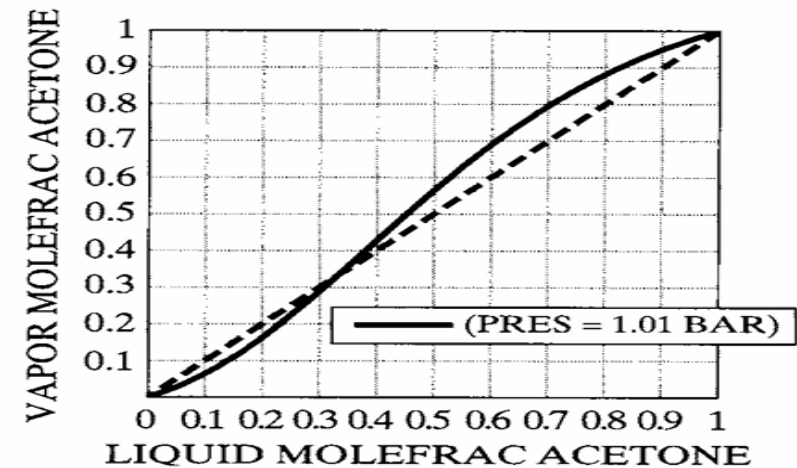
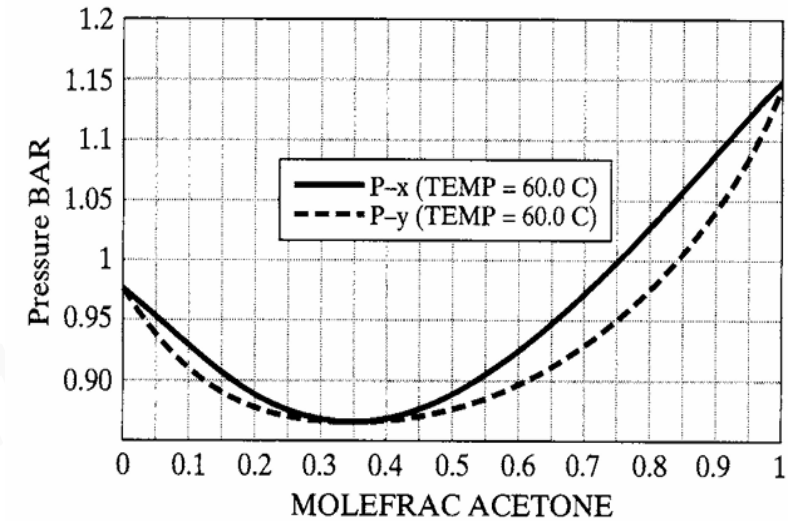
Positive Deviation

Minimum boiling azeotrope
Isopropyl ether-isopropyl alcohol at 70 °C



Negative Deviation

Maximum boiling azeotrope
Acetone-chloroform at 60 °C



Residue Curves

- To **understand the properties of azeotropic mixtures** that contain three chemical species, examine the **properties of residue curves on a ternary diagram**.
- **A collection of residue curves, which is called a residue curve map**, can be computed and drawn by any of the major simulation programs.
- **Each residue curve is constructed by tracing the composition of the equilibrium liquid residue of a simple (Rayleigh batch) distillation in time, starting from a selected initial composition of the charge to the still, using the following numerical procedure.**
- Consider L moles of liquid with mole fractions x_j ($j = 1, \dots, C$), in a simple distillation still at its bubble point. There are no trays and no reflux is provided.
- As heating begins, a small portion of this liquid, ΔL moles, is vaporized.
- The instantaneous vapor phase has mole fractions y_j ($j = 1, \dots, C$), assumed to be in equilibrium with the remaining liquid.
- Since the residual liquid, $L - \Delta L$ moles, has mole fractions $x_j + \Delta x_j$, the mass balance for species j is given by

Residue Curves

$$Lx_j = (\Delta L)y_j + (L - \Delta L)(x_j + \Delta x_j) \quad j = 1, \dots, C - 1$$

In the limit, as $\Delta L \rightarrow 0$,

where

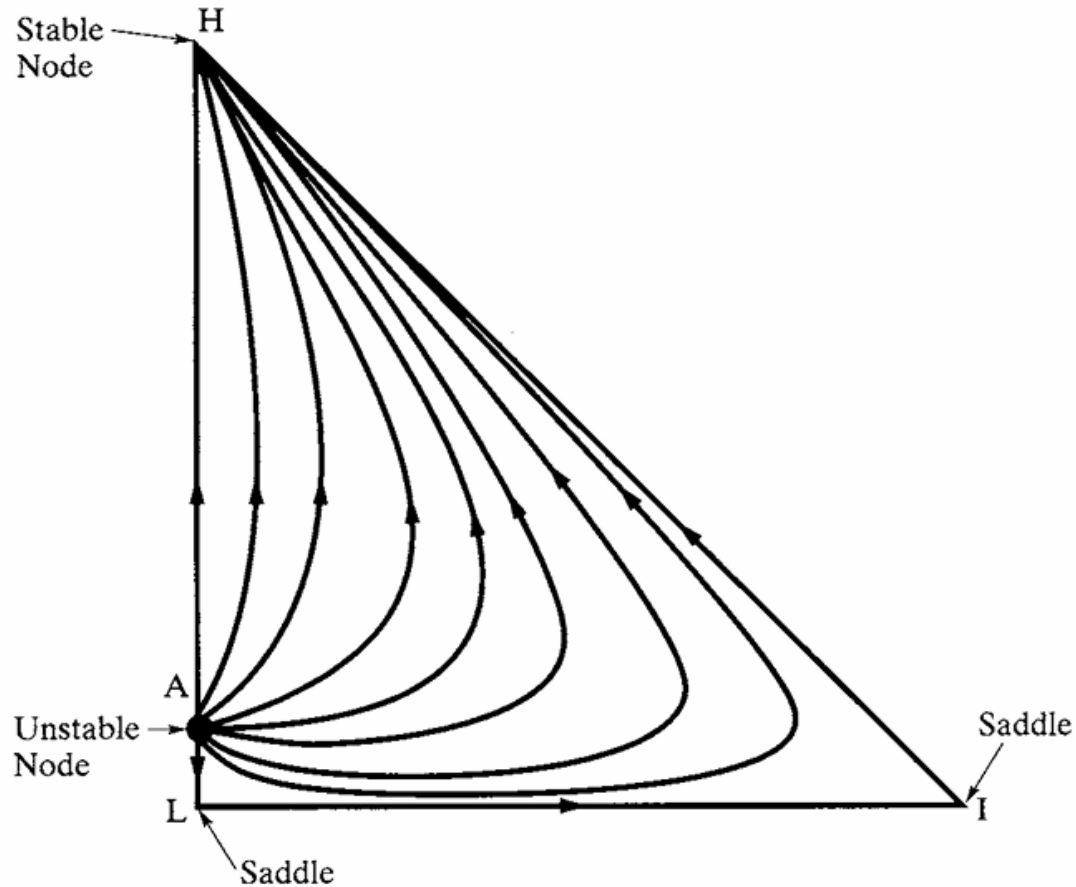
$$\frac{dx_j}{dL/L} = x_j - y_j = x_j (1 - K_j \{T, P, \underline{x}, \underline{y}\}) \quad j = 1, \dots, C - 1 \quad K_j = \frac{y_j}{x_j} = \frac{\gamma_j^L f_j^L}{\phi_j^V P} = 1$$

and setting, $d\hat{t} = dL/L$,

$$\boxed{\frac{dx_j}{d\hat{t}} = x_j - y_j} \quad j = 1, \dots, C - 1$$

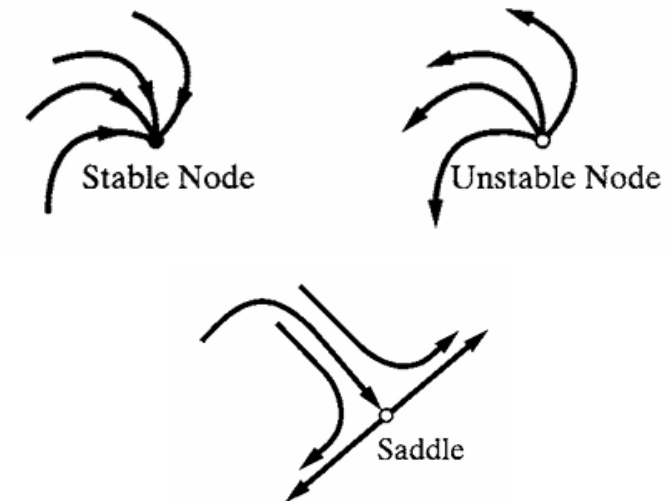
- $\hat{t}$ can be interpreted as the dimensionless time, with the solution defining a family of residue curves.
- **Each residue curve is the locus of the compositions of the residual liquid in time, as vapor is boiled off from a simple distillation still.**
- An arrow is assigned in the direction of increasing time (and increasing temperature).
- For a ternary system, the stability can be determined by calculating the eigenvalues of the Jacobian matrix of the nonlinear ordinary differential equations.

Residue Curves

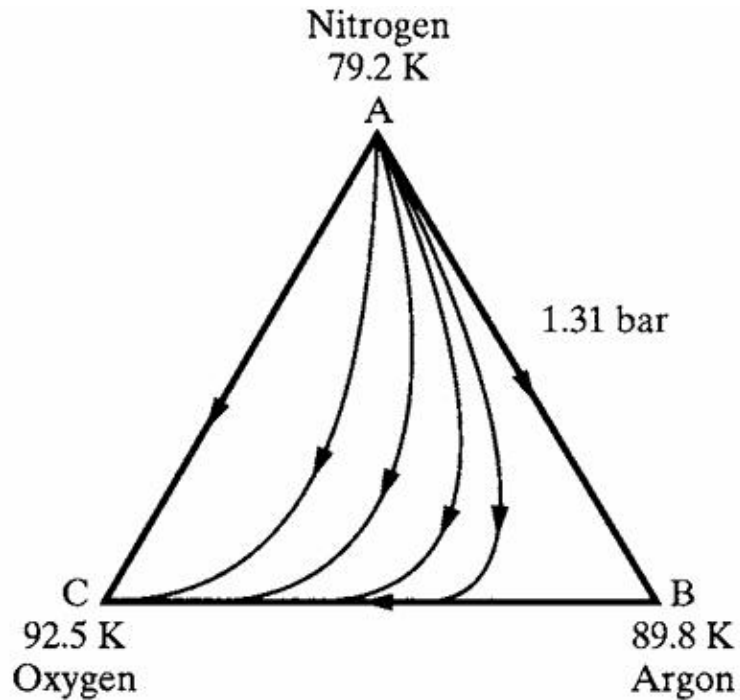


Residue curves of a ternary system with a minimum boiling binary azeotrope

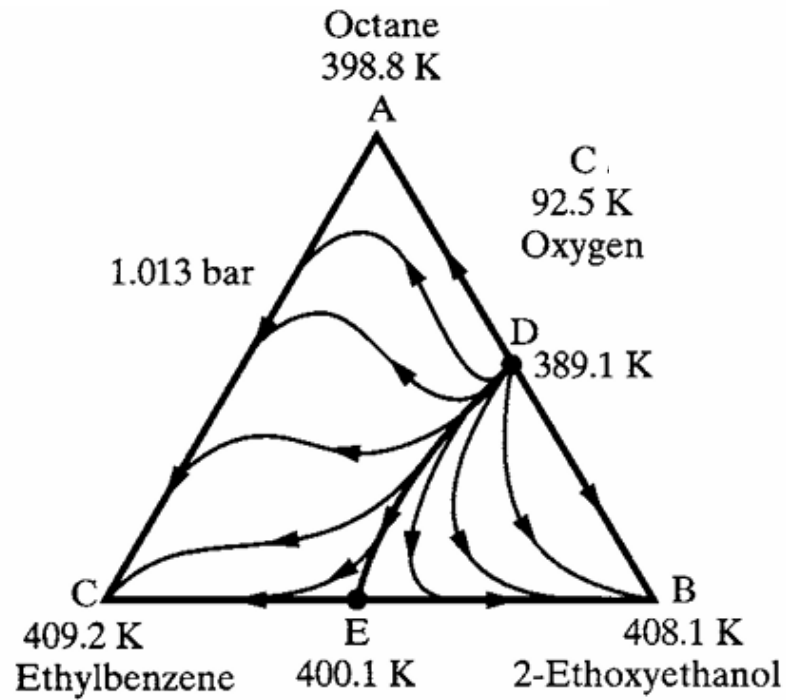
- Fixed points of residue curves: points where driving force for a change in liquid composition is zero $dx/d\hat{t} = 0$.
- **Fixed points are azeotropic points and pure species vertices.**
- **Stable node:** when all of the residue curves are directed by the arrows to the fixed point.
- **Unstable node:** when all are directed away.
- **Saddle point:** when some of the residue curves are directed to and others are directed away from the fixed point.



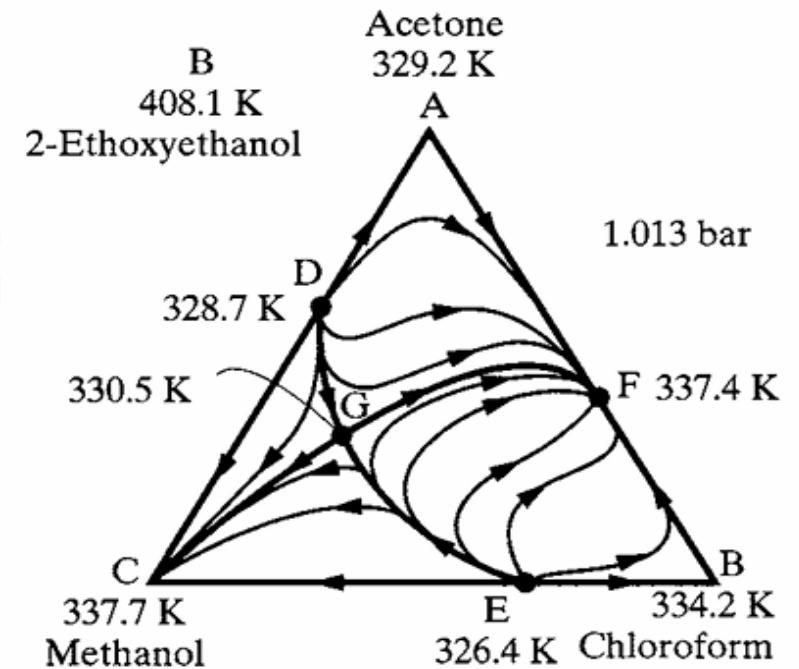
Maps of Residue Curves



System without azeotropes



System with two binary azeotropes



System with binary and ternary azeotropes

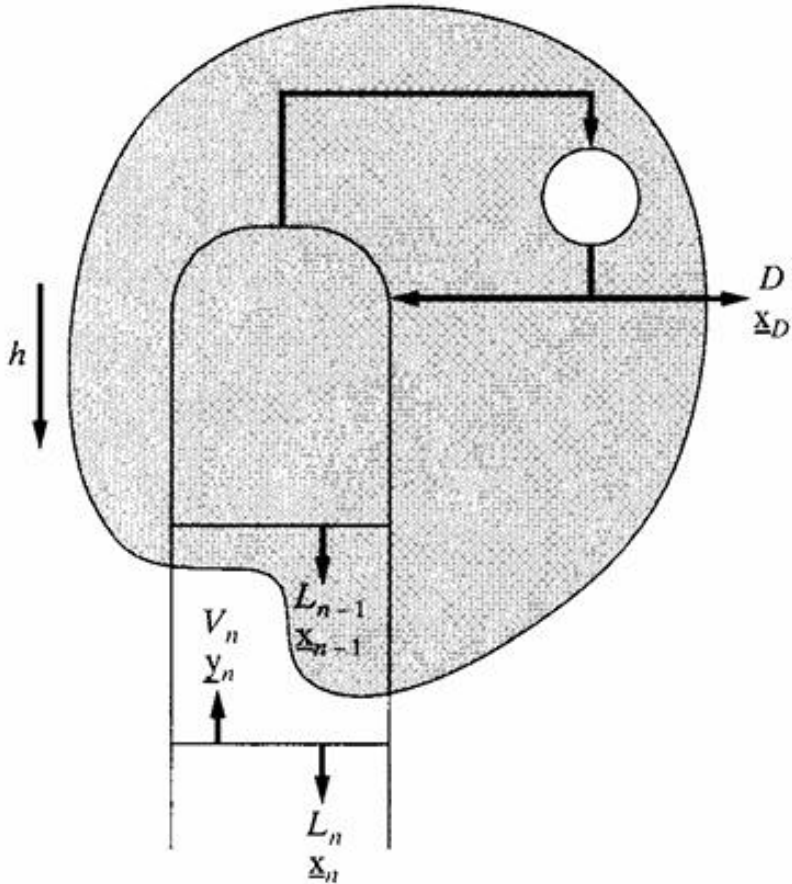
Maps of Residue Curves

- One or more **simple distillation boundaries called separatrices** (curved line DE in (b)) divide these diagrams into regions with distinct pairs of starting and terminal points.
- For the separation of homogeneous mixtures by simple distillation, these separatrices cannot be crossed unless they are highly curved.
- A feed located in region ADECA in (b) has a starting point approaching the composition of the binary azeotrope of octane and 2-ethoxyethanol and a terminal point approaching pure ethylbenzene, whereas a feed located in region DBED has a starting point approaching the same binary azeotrope but a terminal point approaching pure 2-ethoxyethanol. In this case, a pure octane product is not possible.
- (c) is even more complex. It shows **four distillation boundaries** (curved lines GC, DG, GF, and EG), which divide the diagram into **four distillation regions**.

Distillation Towers

- When tray towers are modeled assuming vapor-liquid equilibrium at each tray, **the residue curves approximate the liquid composition profiles at total reflux.**

$$L_{n-1}\underline{x}_{n-1} + D\underline{x}_D = V_n\underline{y}_n$$



- Defining h as the dimensionless distance from the top of the tower, a backward-difference approximation at tray n + 1 is

$$\left. \frac{dx}{dh} \right|_{n+1} \cong \underline{x}_n - \underline{x}_{n-1} \quad \left. \frac{dx}{dh} \right|_{n+1} \cong \underline{x}_n - \frac{V_n}{L_{n-1}} \underline{y}_n + \frac{D}{L_{n-1}} \underline{y}_D$$

- At total reflux, with $D = 0$ and $V_n = L_{n-1}$

$$\left. \frac{dx}{dh} \right|_{n+1} \cong \underline{x}_n - \underline{y}_n$$

- Equation approximates the operating lines at total reflux, residue curves approximate operating lines of a distillation tower operating at total reflux – is known as a distillation line.

Distillation line boundaries and Feasible Product Compositions

- At total reflux, x_B and y_D reside on a distillation line, lie collinear with the feed composition, x_F on the overall material balance line.
- As the number of stages increases, the operating curve becomes more convex and in the limit approaches the two sides of the triangle that meet at the intermediate boiler.
- When a distillation tower operates with a partial condenser, as the feed and product streams are decreased toward total reflux, the last bubble of vapor distillate has the mole fractions, y_D (A and B).
- As total reflux is approached, material balance line connecting the bottoms, feed, and distillate mole fractions is shown.
- **(A) shows distillation line that passes through the x_B and y_D mole fractions, while (B) shows the residue curve that passes through x_D mole fractions, and approximately through x_B mole fractions.**

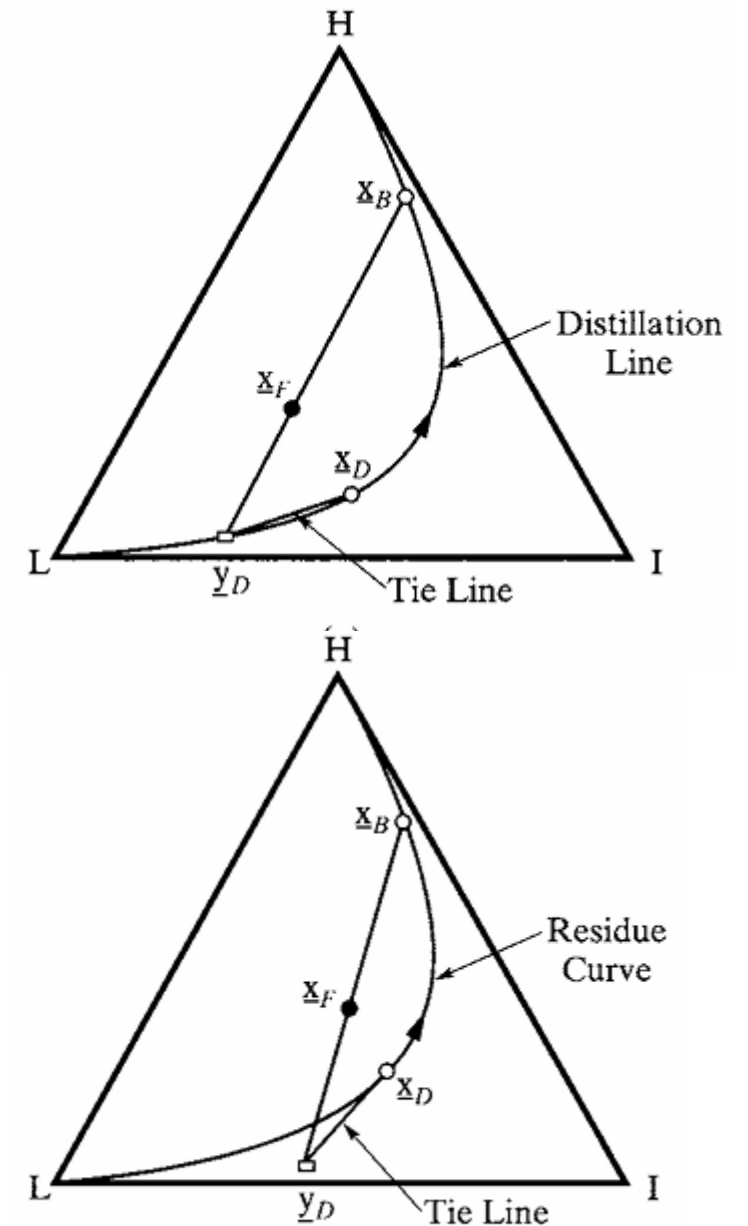
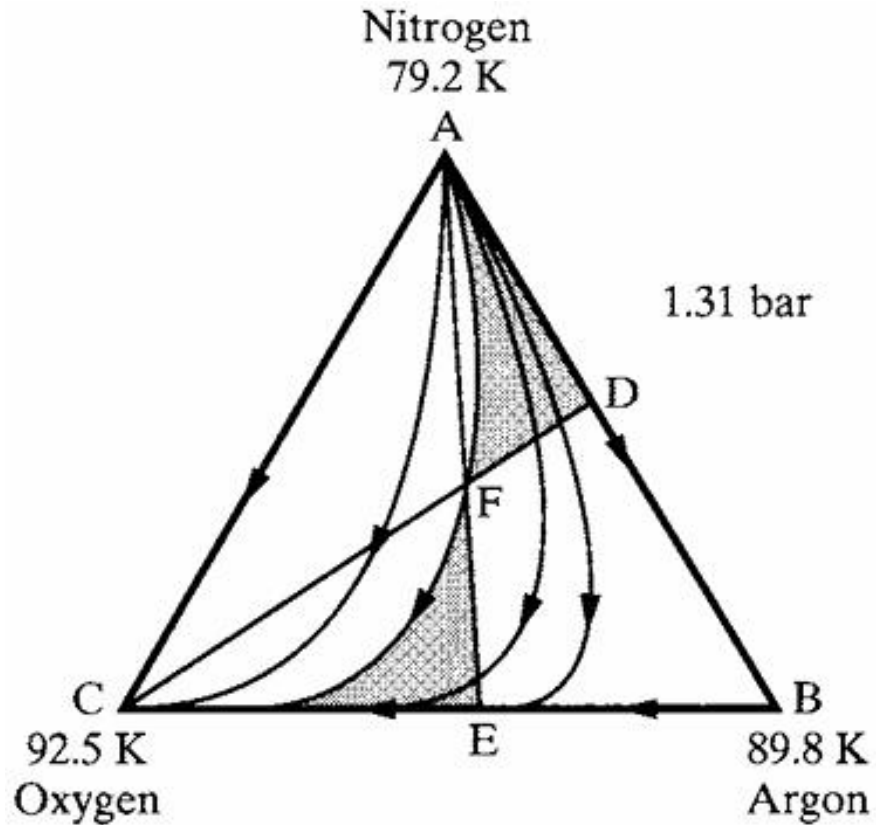
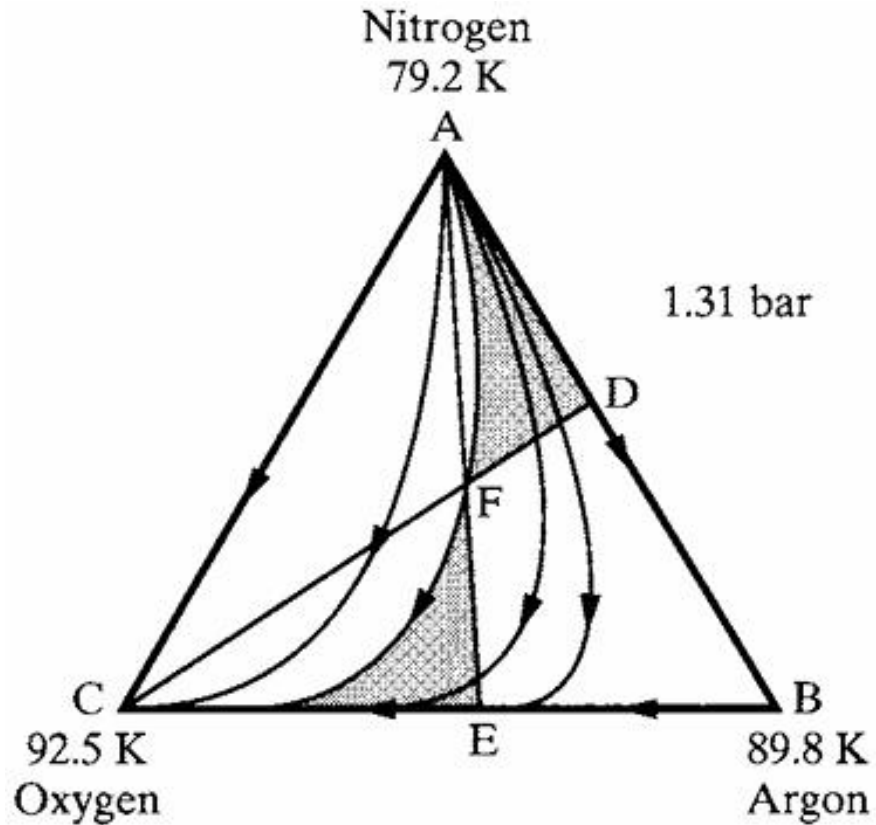


Figure 7.30 Overall mass balance line with a partial/total condenser.



Regions of feasible distillate and bottoms product compositions (shaded) for a ternary mixture: (a) system without azeotropes.

- An operating line at total reflux (minimum stages) is the curve AFC in (a).
- At the other extreme, as the number of stages increases, the operating curve becomes more convex approaching ABC, where the number of stages approaches infinity (corresponding to minimum reflux).
- Hence, **the operating line for a distillation tower that operates within these limiting regimes lies within the region ABCFA.**
- Two additional bounds are obtained. First, in the limit of a pure nitrogen distillate, the **line AFE represents a limiting overall material balance for a feed composition at point F**, with point E at the minimum concentration of oxygen in the bottoms product.
- In the limit of a pure oxygen bottoms, the **line CFD represents a limiting overall material balance**, with point D at the minimum concentration of nitrogen in the distillate along the nitrogen-argon axis.



Regions of feasible distillate and bottoms product compositions (shaded) for a ternary mixture: (a) system without azeotropes.

- Hence, the **distillate composition is confined to the shaded region ADFA**, and the **bottoms product composition lies in the shaded region, CEFC**.
- Operating lines that lie within the region ABCFA connect the distillate and bottoms product compositions in these shaded regions.
- **At best, only one pure species can be obtained.**
- In addition, only those species located at the end points of the distillation lines can be recovered in high purity, with one exception to be noted.
- ***Hence, the end points of the distillation lines determine the potential distillate and bottoms products for a given feed.***
- ***The location of the feed point determines the distillation region in which the potential distillate and bottoms product compositions lie.***

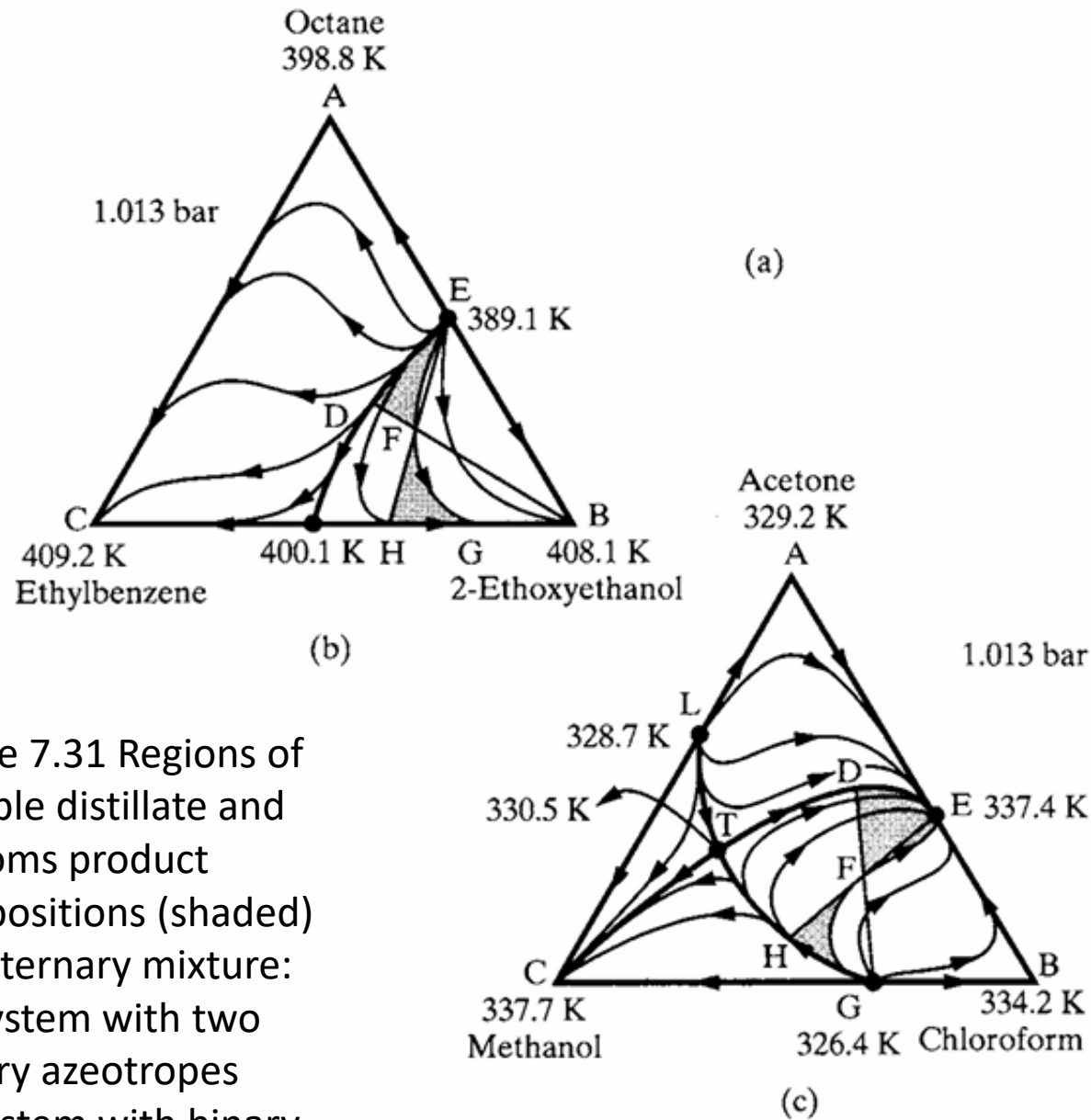


Figure 7.31 Regions of feasible distillate and bottoms product compositions (shaded) for a ternary mixture: (b) system with two binary azeotropes (c) system with binary and ternary azeotropes (Stichlmair et al., 1989).

- Figure 7.3 b, for feed F, only pure 2-ethoxyethanol can be obtained.
- When the feed is moved to the left across the distillation-line boundary, pure ethylbenzene can be obtained.
- In Figure 7.3 c, only methanol can be recovered in high purity for feeds in the region LTGCL. For a feed in the region EDTHGBE, no pure product is possible.
- Before attempting rigorous distillation calculations with a simulation program, it is essential to establish, with the aid of computer-generated residual curve maps, regions of product-composition feasibility.

Quick Recap

- Draw a flowsheet for a process with 1 reactor system.
- List possible methods of phase separation of reactor effluent.
- Write any 4 industrial separation methods with separation agent and separation property.
- List some criteria for selection of separation methods.
- List several equipment available for absorption, adsorption, crystallization, liquid-liquid extraction.
- Write equation for the no. of possible sequences of ordinary distillation columns.
- What is marginal vapor rate method?
- What is an azeotrope? What are its types?
- Write equation of residue curve and distillation line, draw a residue map with distillation line.